Computational Sustainability: Computing for a Sustainable Future

Carla P. Gomes
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Institute for Computational Sustainability
Department of Computer Science
Cornell University

AAAI Symposium

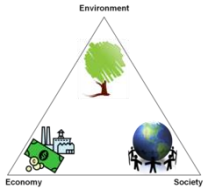
Plan4Planet
November 2026

Computational Sustainability

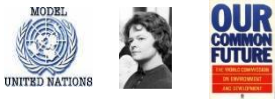
Interdisciplinary field that aims to harness the power of AI and computational methods for Sustainable Development.

Sustainable Development encompasses balancing

- *environmental,*
- *economic, and*
- *societal needs.*



United Nations, Our Common Future, 1987



Gro Brundtland

Ultimate goal of Sustainable Development

HUMAN WELL-BEING

of **current and future generations.**



NSF Expeditions in Computing (CISE)

Thank you!

Expeditions in Computing (CISE)

1st Expeditions in Computational Sustainability (2008)



Conference, Referred CompSust Tracks, Workshops

- **To nucleate** the Computational Sustainability field
- **To identify** a number of **core research directions** for **maximal impact**, both in terms of **Computer Science** and **Sustainability.**

2nd Expeditions: Large-Scale Research Network for Expanding the Horizons of Computational Sustainability [2016]

Cornell University, Bowdoin, Caltech, Carnegie Mellon University, Georgia Tech, Howard University, Harvard University, The Ohio State University, Oregon State University, Princeton University, Stanford University, UMASS Amherst, USC University of Southern California, Vanderbilt University

and several international universities, as well as Gov. organizations and NGOs.



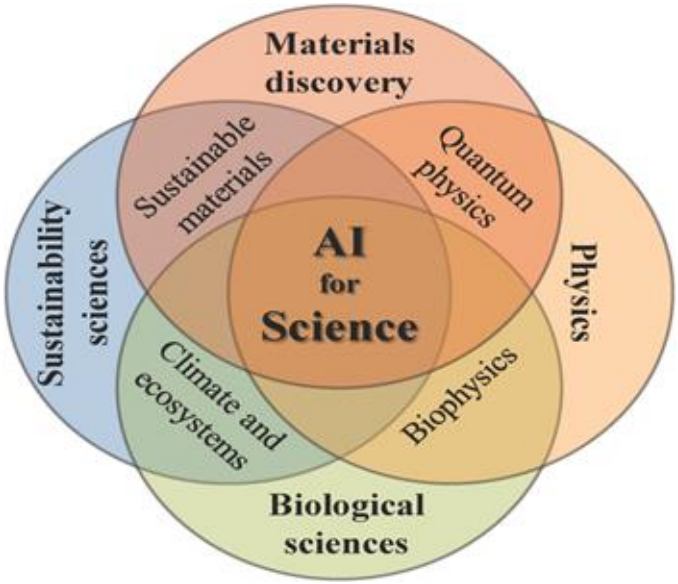


CompSustNet

300+ faculty, students, and collaborators!!!

CU AI Sci

The Cornell University AI for Science Institute



Schmidt Sciences

Eric and Wendy Schmidt
AI in Science Postdoctoral Fellowship,
a program of SCHMIDT FUTURES

- 100 Postdocs
- Over 6 years

AI-LEAF

NSF/NIFA AI Institute
Land, Economy, Agriculture and Forestry

NSF NRT

Cornell AI4S Initiative
Artificial Intelligence for Sustainability

Thank you!

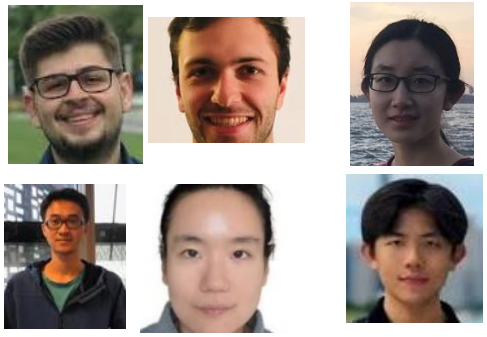
Thank you!

My PhD students and postdocs

(connections to computational sustainability and scientific discovery)

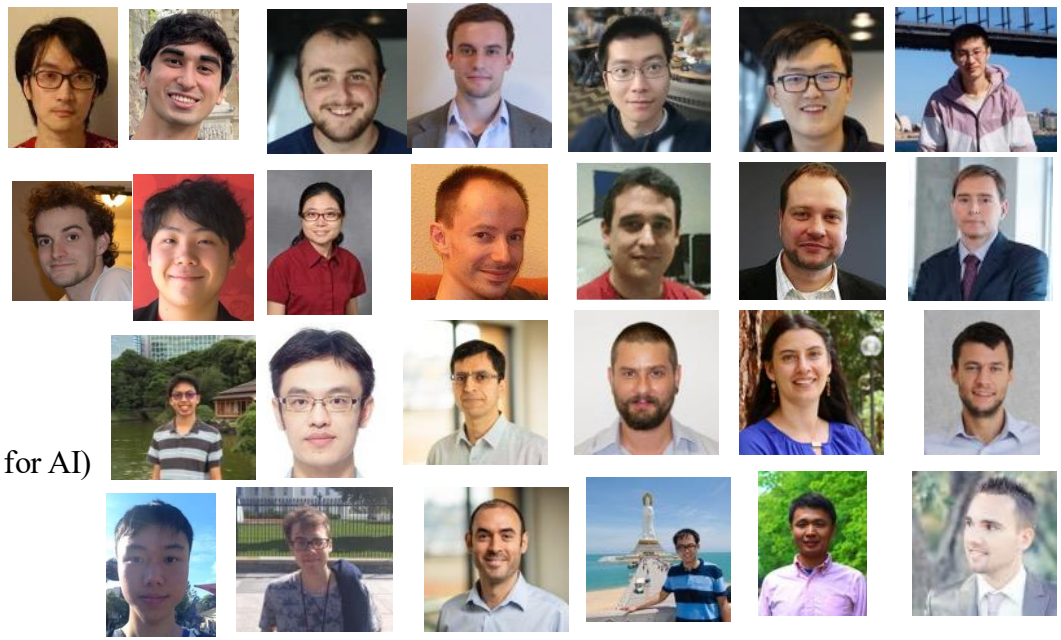
Current:

Utku Acikalin (Phd, CS)
Christian Belardi (PhD CS)
Joshua Fan (PhD, CS)
Victoria Mao (PhD, CS)
Shufeng Kong (Postdoc, CS)
Delia Qu (PhD, CS)



Past postdocs and PhD students:

Sebastian Ament (Meta)
Carlos Ansotegui (professor, Univ. Lleida, Spain)
Junwen Bai (Google)
Yiwei Bai (DRAI)
Ramon Bejar (professor, Univ. Lleida, Spain)
Johan Bjorck (Microsoft)
Di Chen (DRAI)
Theo Damoulas (professor, University of Warwick, UK)
Bistra Dilkina (associate professor, USC)
Yuanqi Du ((PhD, CS)
Stefano Ermon (associate professor, Stanford University)
Dieqiao Feng (PhD, CS)
Aaron Ferber (postdoc)
Cesar Fernandez (professor, Univ. Lleida, Spain)
Marc Grimson (PhD, CS)
Yunsong Guo (Google)
Ronan Le Bras (research scientist, P.Allen Inst. for AI)
Yimeng Min (PhD, CS)
Guillaume Perez (Ugmented)
Brendan Rappazzo (Morgan Stankkey)
Ashish Sabharwal (senior research scientist, P. Allen Inst. for AI)
Meinolf Sellmann (Shopify)
Qinru Shi (Baidu)
Willem van Hoeve (professor, CMU)
Xiaojian Wu (research scientist, Facebook)
Yexiang Xue (associate professor, Purdue University)



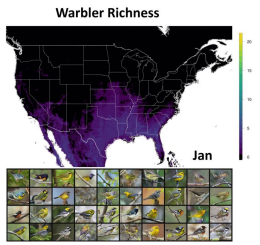
Computational Sustainability Examples

1 - AI to combat the rapid decline in biodiversity

(e.g., fine-grained joint species distributions, strategic planning of wildlife corridors)

Dilkina et al. *Conservation Biology 2016* A. Gupta et al. *Conservation Biology 2019.*, Davis et al. *Ecology 2023*

CS papers in CP, AAAI, IJCAI, HCOMP, ICML, NeurIPs



Bird Conservation



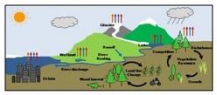
Wildlife corridors



Global Freshwater Fish Conservation

2 - AI to predict and understand the soil organic carbon cycle

Xu al. Geosci. Model Dev. 2026. *Fan et al AAAI 26*



Soil Carbon Cycle

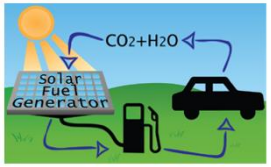
3- AI to accelerate materials discovery and metabolite discovery

Gomes et al. *Nature Rev. Materials 2021.* Wang et al *Nature 2023*

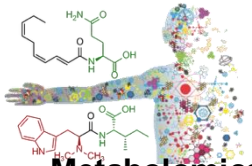
Chen, et al. *Nature Machine Intelligence,* September 2021.. Ament et al. *Science Advances, 2021.*

Kong et al. *Nature Communications 2022*

CS papers in CP, AAAI, IJCAI, HCOMP, ICML, NIPs



Solar fuels



Metabolomics



Rio Santiago
(free-flowing Andean Amazon river with large planned hydropower dams)



200+~400 proposed hydropower dams Amazon basin

4 - AI for strategic planning of hydropower expansion to reduce adverse impacts on people and nature

(200+ ~400 new) (ecosystem impacts and tradeoffs of hydropower



Almeida et al. *Nature Communications 2018* A. Flecker et al. *Science 2022.*

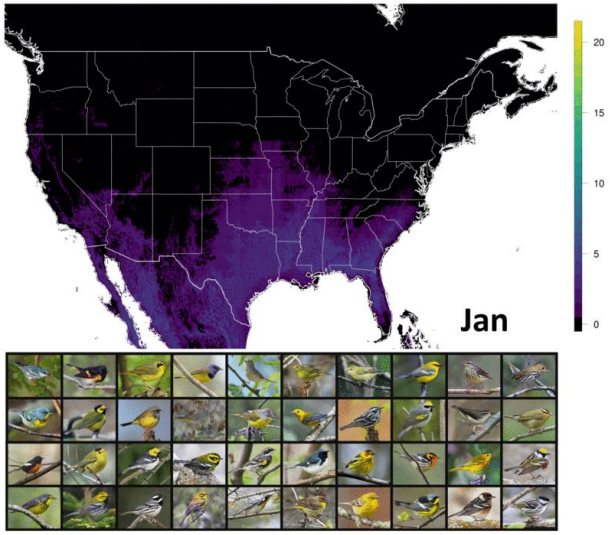
Pacheco et al. *Nature Sustainability 2025* *Heilpern et al. Nature Sustainability 2025*

CS papers CPAIOR, AAAI, IJCAI, AAMA, NeurIPS

→ **Decision Making Systems**: considering multiple objectives (Pareto optimization rather than standard single objective optimizatio

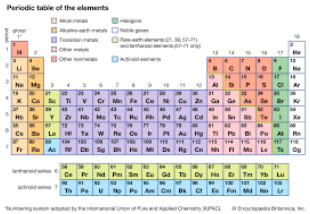
Transferability of AI:
From bird species to chemical species

Warbler Richness

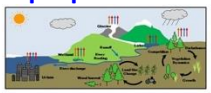


Deep Models
Joint bird distributions

Prediction Tasks

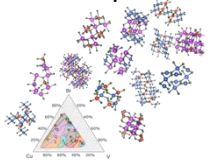


Joint properties of Materials

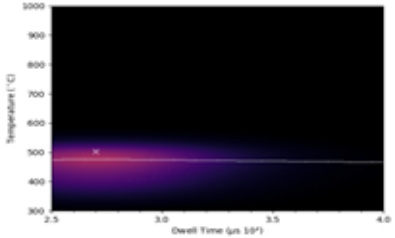


Soil Organic Carbon Cycle

Unsupervised Inverse Problems
(De-Mixing or Source Separation)

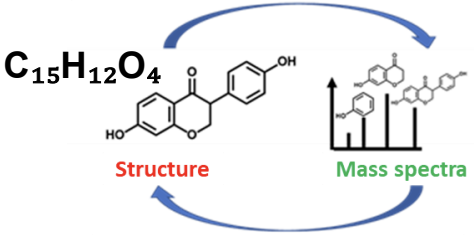


Bi-Cu-V-O Crystal-phase map



Bi_2O_3 Crystal-phase boundary map

***Forward**: predict spectra of known metabolite based on its structure*



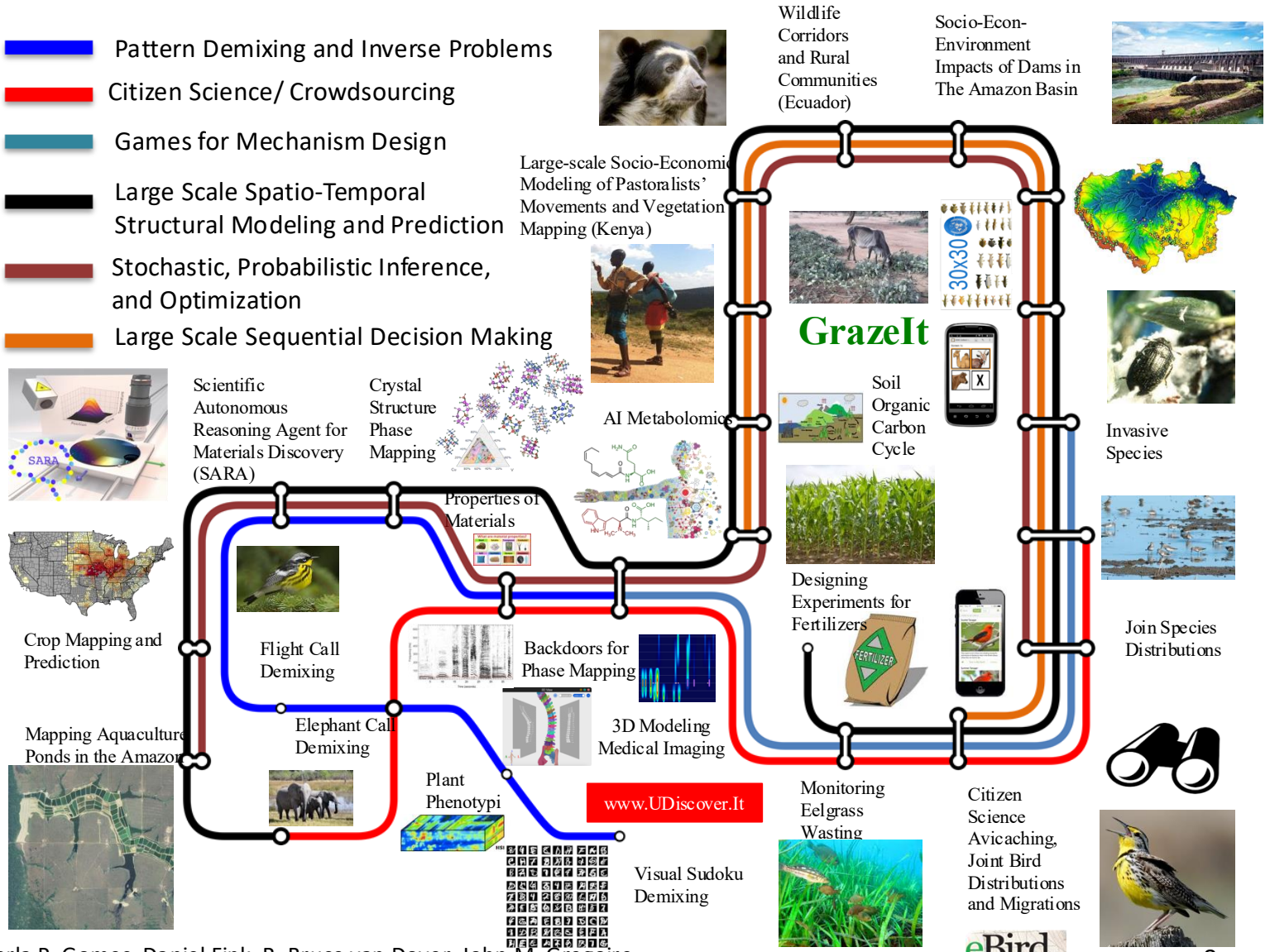
***Inverse**: predict structure of unknown metabolite from spectra*

Computational models for scientific discovery

nature REVIEWS MATERIALS

Computational Sustainability Meets Materials Science
Gomes et al. *Nature Rev. Materials 2021*

Transferability of AI Methodologies across domains:
Cross-cutting computational themes
Subway Lines



nature REVIEWS MATERIALS

Carla P. Gomes, Daniel Fink, R. Bruce van Dover, John M. Gregoire.

Computational sustainability meets materials science. *Nature Reviews Materials* (2021).

Knowledge-Centric AI:
From Data-Driven AI to Knowledge-Centric AI

Over the last decade --- tremendous progress in AI → Artificial General Intelligence (AGI)
Focus has been on: Data-Driven AI (Deep Learning, LLMs, etc.).

The Gap:

Pure data-centric approaches have inherent limitations for scientific discovery and decision-making.

→ **Scientific Artificial Intelligence (SAI)** *(beyond "general" intelligence).*

Approach: *Inject* ***Knowledge-Centric Reasoning into Data-Driven AI.***

Goal: Enable AI systems to reason from first principles, prior knowledge, and data

- *To deal with data scarcity and lack of labeled data.*
- *To generalize beyond training distribution.*
- *To respect hard constraints and physical laws.*
- *To deal with the complexity of large combinatorial search spaces*
- *To consider multi-objectives and Pareto frontiers instead single objective approaches*

More Interpretable Knowledge-Centric Architectures

Deep Neural Networks and Transformers:

- Highly complex (non-linear) functions
- Black-boxes

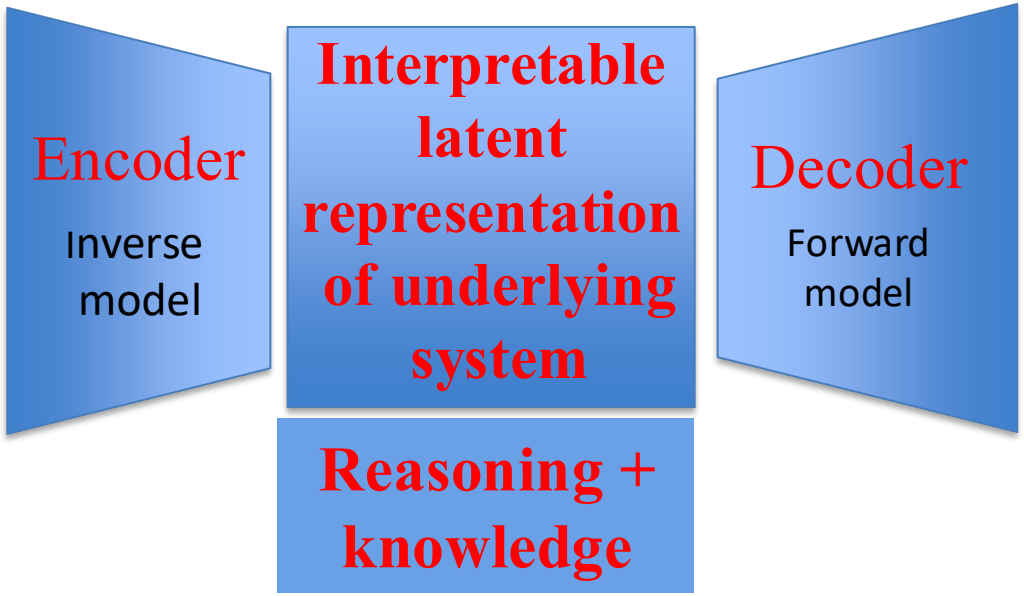
(multiple non-interpretable hidden layers)

- Rely on large amounts of training data

Chat GPT 3 - **175 billion parameters**
Chat GPT now - **>1Trillion** parameters
Kimi K3 (Moonshot AI (Alibaba)
→ **2.8 Trillion** parameters

Key Requirements of
Scientific Models
Interpretability
and
Adherence to Scientific Principles
and
Deal with Scarce Labeled Data

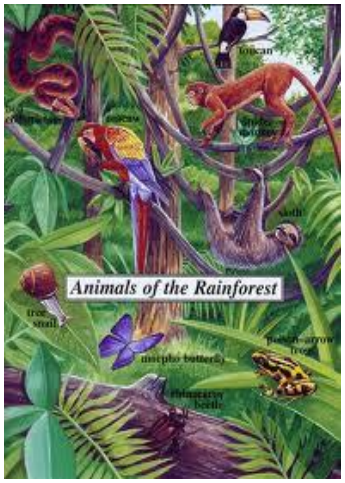
Our overall approach
Open-up the black box
More Interpretable
Knowledge-Centric Architectures



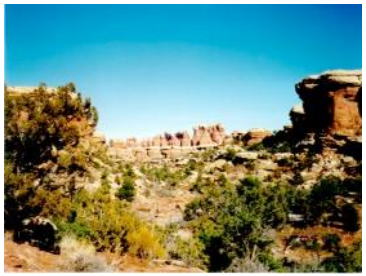
AI to combat the rapid biodiversity decline in biodiversity



Biodiversity or Biological Diversity







Fundamental question in biodiversity research:
How different species are distributed across landscapes over time.

Particular important given the tremendous increasing rate of biodiversity reduction since the Industrial Revolution due to habitat fragmentation and climate change.

Variety of Sensor Technologies

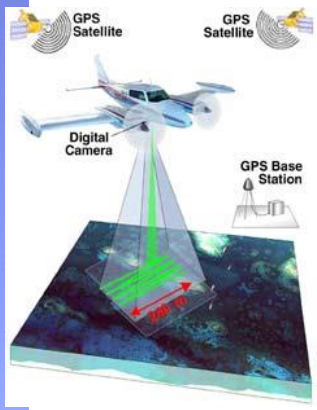


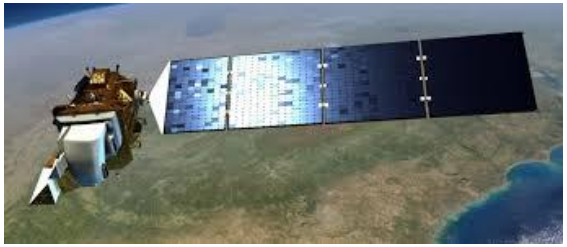




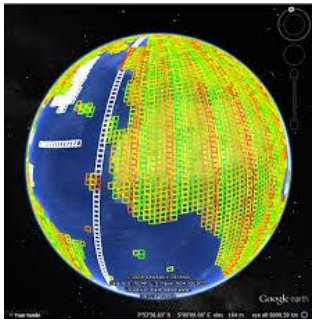


Photo courtesy of www.carboafrica.net

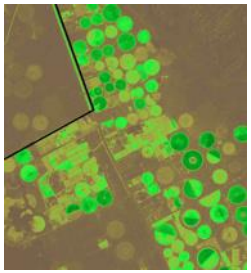


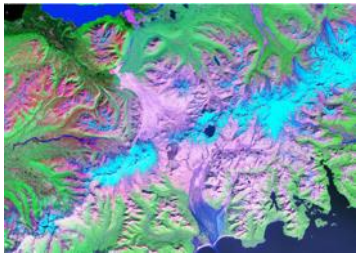


LandSat
~50years old









LandSat
images



Very sophisticated sensor

Citizen Science and Species distributions

eBird
Citizen Science



1 Million+ volunteer birders

Bird Observations

1 Billion+ bird observations

>5X



Distribution Models for 500+ species with weekly estimates at fine spatial resolution (3km²)

State of the Birds Report
(officially released by Secretary of Interior)

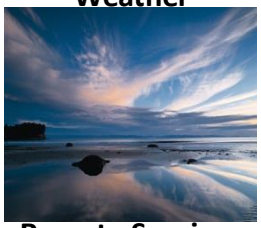


Novel Approaches To Conservation Based on eBird Models

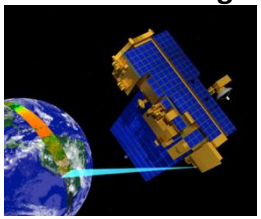
Land Cover



Weather

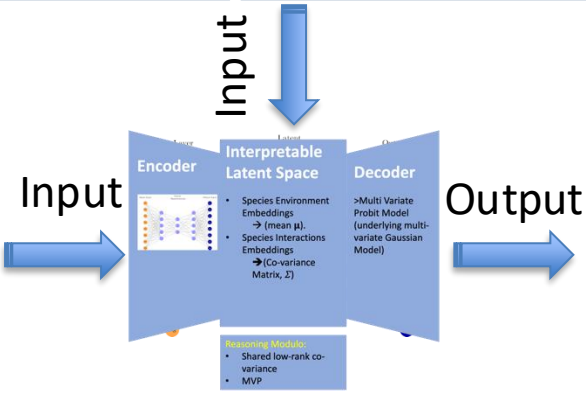


Remote Sensing



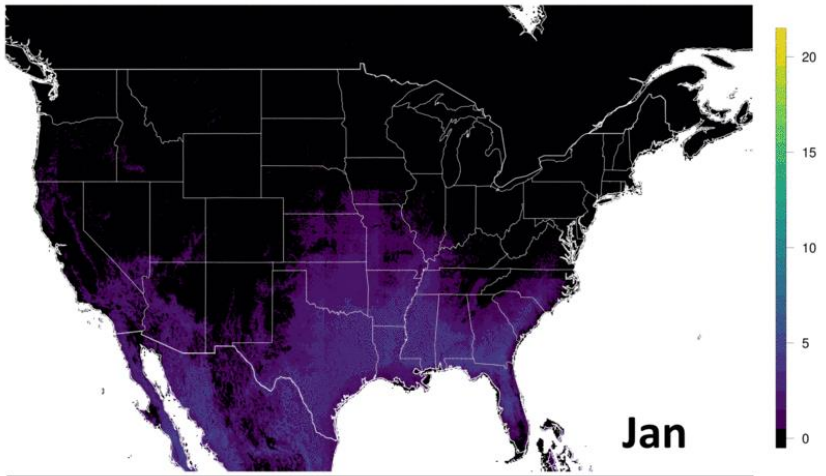
Environmental Data

Input



Spatial and Temporal Deep Reasoning Models

Warbler Richness (40 Species)





Joint Migration Pattern for 40 Species of Warblers

The models reveal the habitat preferences and migrations of the birds, at a fine resolution.

Dynamic High-Precision Conservation
Bird Returns

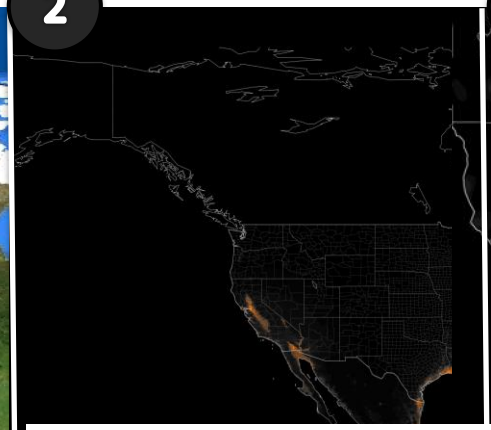
Protecting Migratory WaterBirds in California Against Drought

1



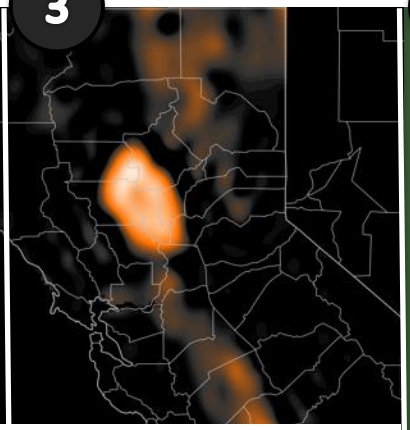
Pacific Migration Flyway

2



eBird Models

3



Target Estimates

4

Farmers Submit Bids

The Nature Conservancy

Bids selected based on targeted estimates

Reverse Auction Bid Selection

The Nature Conservancy has generated over 20,000 acres of additional habitat for waterbirds in California

Combinatorial Reverse Auctions

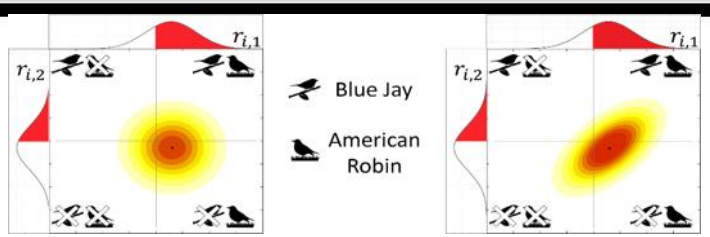


Sacramento Valley, CA

Farmers submit bids to TNC to keep the target rice fields flooded during the short periods of bird migration in California.

Radically novel, dynamic, way of doing bird conservation. Only possible because of advanced computational methods.

Beyond Single Species Models: Knowledge-Centric Joint Species Distribution Models (JSDM)



- **Species dependencies**

Competition, cooperation, etc.
(not captured in most previous models)

- **Ecological foundations for presence-absence JSDMs:**

Important to capture:

Preferences of species wrt environment

Interactions among species

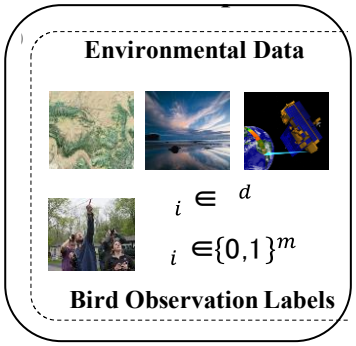
Multivariate Probit Model (presence-absence model)

Species are governed by an underlying multi-variate Gaussian model

- **Limitations of state-of-the-art MVP models:**
  - Features are manually engineered
  - Challenging to train MVP model assumes an underlying multivariate Gaussian distribution of species interactions – hard to estimate parameters of co-variance matrix

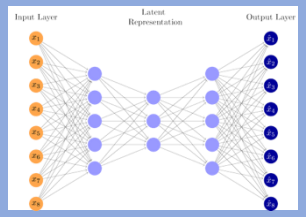
Deep Multivariate Probit for Joint Species Distribution: Incorporate Ecological Insights into Deep Learning (Supervised Learning)

Input



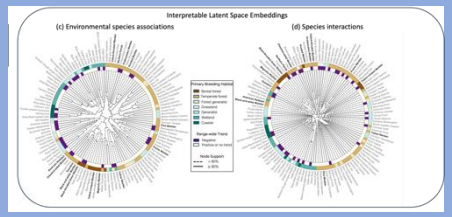
Citizen Science Data Input

Encoder



Interpretable Latent Space

- Species Environment Embeddings → (mean $\boldsymbol{\mu}$).
- Species Interactions Embeddings ➔(Co-variance Matrix, Σ)



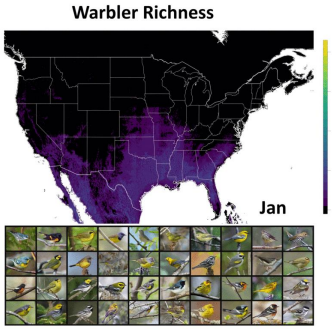
Reasoning Modulo:

- Shared low-rank co-variance
- MVP

Decoder

>Multi Variate Probit Model (underlying multi-variate Gaussian Model) Fully differentable

Output

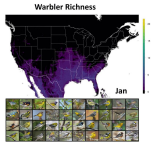


Joint species distribution (eBird)

Loss Function: Negative Joint Log-Likelihood

[Chen, Xue, Gomes, ICML 18] [Davis el al, Ecology 23]

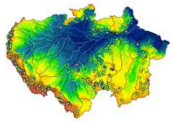
General Framework for Multi-Entity Learning (Classification and Regression): Transferability of AI



➢ **Mapping** species distributions across space and time



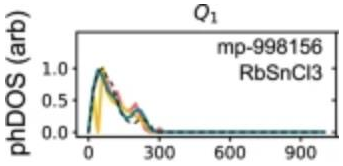
➢ Multi-object detection for computer vision



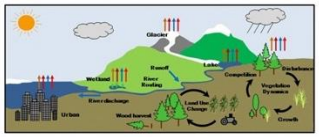
➢ Mapping of **Amazon land cover**



➢ Prediction of properties of materials



➢ Prediction of density of states for materials discovery



➢ Predicting and Understanding soil organic carbon cycle

nature REVIEWS MATERIALS

nature COMMUNICATIONS

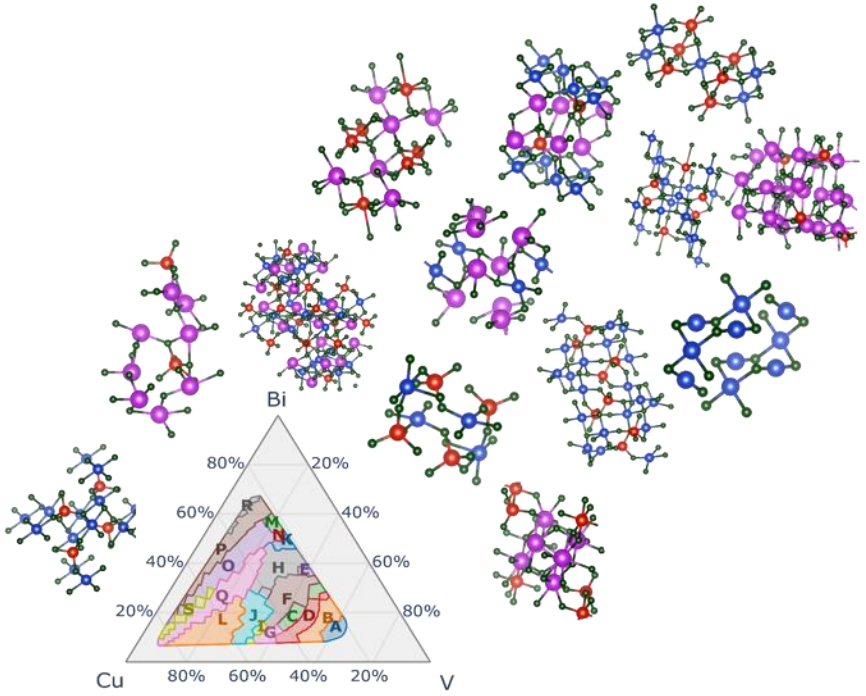
esa ECOLOGY

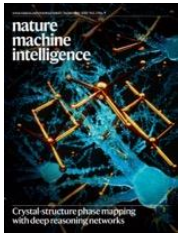
Applied Physics Reviews

ICML / AAAI / NeurIPS

AI to accelerate Scientific Discovery

AI Topic - Inverse Problems:
Unsupervised Pattern De-Mixing





Di Chen, Yiwei Bai, Sebastian Ament, Wenting Zhao, Dan Guevarra, Lan Zhou, Bart Selman, R. Bruce van Dover, John M. Gregoire, Carla P. Gomes. *Nature Machine Intelligence, September 2021. [pre-print in arXiv].*

Can you make sense of these symbols?



Prior knowledge

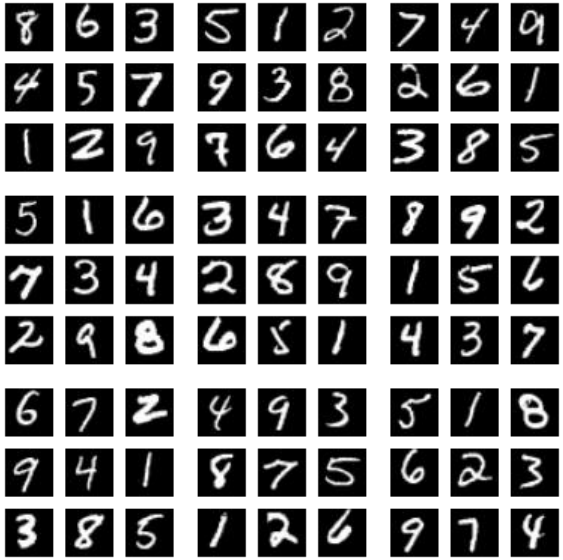
Two overlapping Sudokus:

One using digits 1,2,3,…9

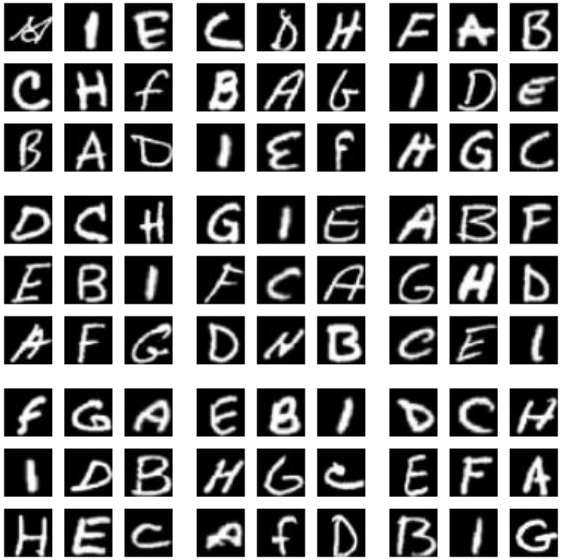
The other using capital letters: A, B, C, D…I

Sudoku rules

(No training Sudoku data)







Deep Reasoning Nets
de-mix the Sudokus
by combining deep learning
with reasoning
about prior knowledge
(Sudoku rules).

NO training Sudoku data.

Deep Reasoning Networks

Combining deep learning with reasoning for discovery:

An intuitive illustrative example

Deep Reasoning Nets de-mix the Sudokus by combining deep learning with reasoning about prior knowledge (Sudoku rules) via an interpretable latent space (capturing digit probabilities and shape embeddings) .

NO Mixed Sudoku training data.

Can you make sense of these symbols?

DRNets **significantly outperform supervised models without reasoning about Sudoku rules**

Prior knowledge

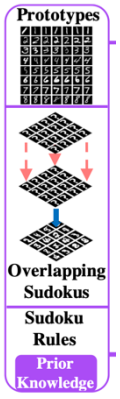
Two overlapping Sudokus:

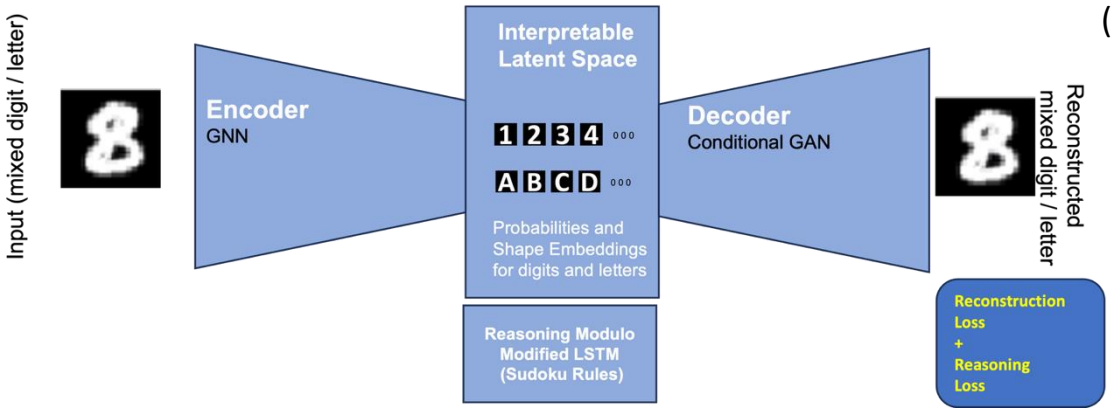
One using digits 1,2,3,…9

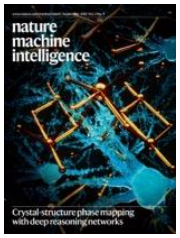
The other using capital lette A, B, C, D…I

Sudoku rules

(No training Sudoku data)







Di Chen, Yiwei Bai, Sebastian Ament, Wenting Zhao, Dan Guevarra, Lan Zhou, Bart Selman, R. Bruce van Dover, John M. Gregoire, Carla P. Gomes. **Automating crystal-structure phase mapping by combining deep learning with constraint reasoning.** *Nature Machine Intelligence, September 2021. [pre-print in arXiv].*

+ ICML18, CP19

DRNets:
Data-and-knowledge driven constrained optimization

Formulate unsupervised pattern de-mixing as **data driven constrained optimization:**

Decoder (forward) model - given the learn demixed patterns (in the latent interpretable space) **generates** the mixture of symbols.

Encoder: Learn the **inverse** problem - the demixed patterns (producing the interpretable latent space)

To reconstruct the initial mixture **minimizing the loss** between the two

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}(G(\phi_\theta(\mathbf{x}_i)), \mathbf{x}_i)$$

$$\text{s.t. } \phi_\theta(\mathbf{x}_i) \in \Omega^{\text{local}} \text{ and } (\phi_\theta(\mathbf{x}_1), ..., \phi_\theta(\mathbf{x}_N)) \in \Omega^{\text{global}} \quad (1)$$

$$\text{where } \phi_\theta(\mathbf{x}_i) := (\mathbf{z}_{i,1}, ..., \mathbf{z}_{i,m}, \mathbf{e}_{i,1}, ..., \mathbf{e}_{i,m})$$

Satisfying pattern rules

We use Lagrangian relaxation to approximate equation (1) with an unconstrained optimization problem, i.e.,

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}(G(\phi_\theta(\mathbf{x}_i)), \mathbf{x}_i) + \lambda^l \psi^l(\phi_\theta(\mathbf{x}_i))$$

$$+ \sum_{j=1}^{N_g} \lambda_j^g \psi_j^g(\{\phi_\theta(\mathbf{x}_k) | k \in S_j\})$$

Entropy-based continuous relaxations to capture combinatorial constraints

Cardinality Constraint	**Cardinality Constraint Relaxation**
$e_{i,j} \in \{0,1\}$ j $= 1 \dots 8$ $s.t. \sum_{j=1}^{4} e_{i,j} = 1$ and $\sum_{j=5}^{8} e_{i,j} = 1$	$\min_{\theta} H(P_i) + H(Q_i)$ $= -\sum_{j=1}^{4} P_{i,j} \log P_{i,j} - \sum_{j=1}^{4} Q_{i,j} \log Q_{i,j}$
All-Different Constraint	**All-Different Constraint Relaxation**
For all constrained set S $s.t. \sum_{i \in S} e_{i,j} = 1$ for j $= 1 \dots 8$	For all constrained set S $\max_{\theta} H(\bar{P}_S) + H(\bar{Q}_S)$
k-Sparsity Constraint	**k-Sparsity Constraint Relaxation**
$e_{i,j} \in \{0,1\}$ j $= 1 \dots$ M $s.t. \sum_{j=1}^{M} e_{i,j} \leq k$	$\min_{\theta} max\{H(P_M), c\}$, where $c < \log k$

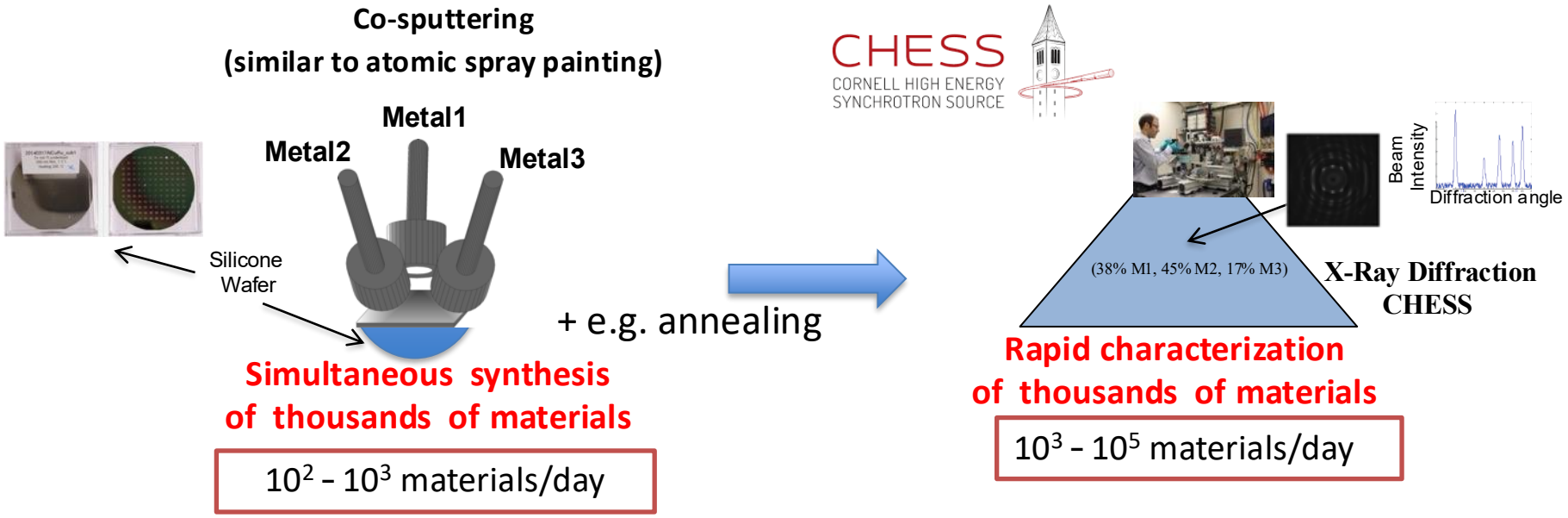
- We propose a group of entropy-based continuous relaxations to encode general discrete constraints such as sparsity, cardinality and All-Different constraints and the constraint-aware SGD (a variant of SGD) to optimize the penalty functions.

Ablation Studies:
Role of Reasoning Module and Conditional Gan

		Methods	Accuracy (%)		Training Set Size	Test Set Size
			Digit	Sudoku		
9x9 Multi-MNIST Sudoku	unsupervised	DRNets	**99.99**	**99.23**		81x10,000 overlapping digits
		DRNets + local search2	**99.99**	**99.84**		
		DRNets w/o Reasoning **(ablation)**	63.82	0.0	N/A	
		DRNets w/o cGAN **(ablation)**	1.24	0.0		
	supervised	CapsuleNet	99.10	23.33	81x10,000 overlapping digits	
		CapsuleNet + local search2	99.10	73.63		
		ResNet-18	98.73	35.39		
		ResNet-18 + local search2	98.73	80.72		

DRNets significantly outperformed the supervised models CapsuleNet, ResNet, and DRNets without reasoning **in Sudoku accuracy (and even performed slightly better in digit accuracy)**.

High-Throughput Materials Discovery: Crystalline Phase Mapping Problem from XRD data



Simultaneous synthesis of thousands of materials

10^2 – 10^3 materials/day

Rapid characterization of thousands of materials

10^3 – 10^5 materials/day

BIG Challenge: High-Throughput Crystal Structure Discovery

How to infer the crystal structure of the materials, based on the X-ray diffraction patterns

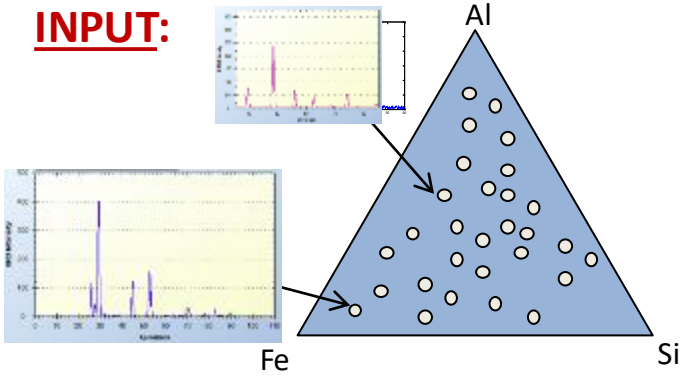
(Phase Map Identification Problem)

Mainly manual task requiring expert knowledge!

Only a few systems a year

Inverse Problem: Crystalline Phase Mapping Problem from XRD data: Unsupervised Pattern De-Mixing

INPUT:



Collection of XRD Patterns

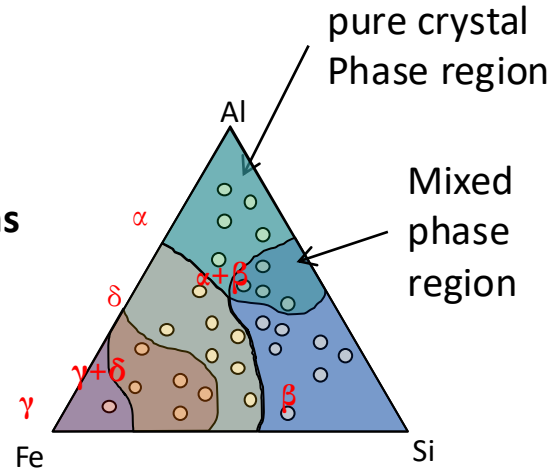
Additional Physical Requirements (thermodynamic rules):

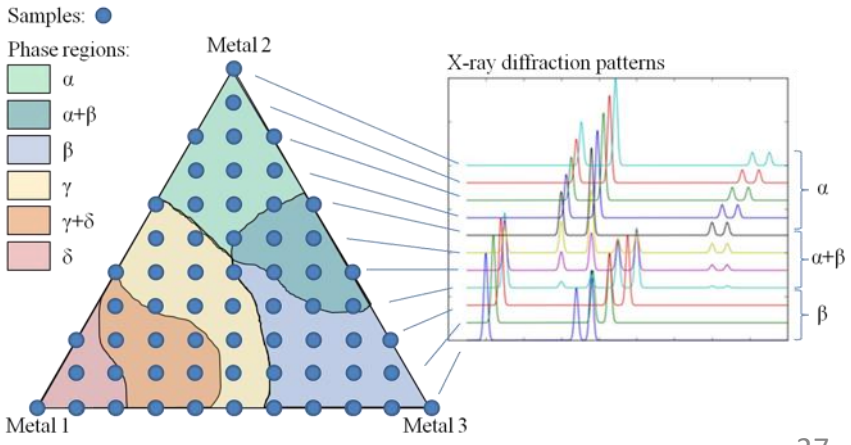
- Phase Connectivity
- Gibbs Rule: # pure phases in a mixture (per sample) <= # elements
- Gibbs alloy: if alloying => -1 degree of freedom
- Peaks shift by ≤ 15% within a region
  - Continuous and Monotonic
- Small peaks might be discriminative
- Peak locations matter, more than peak intensities

OUTPUT:

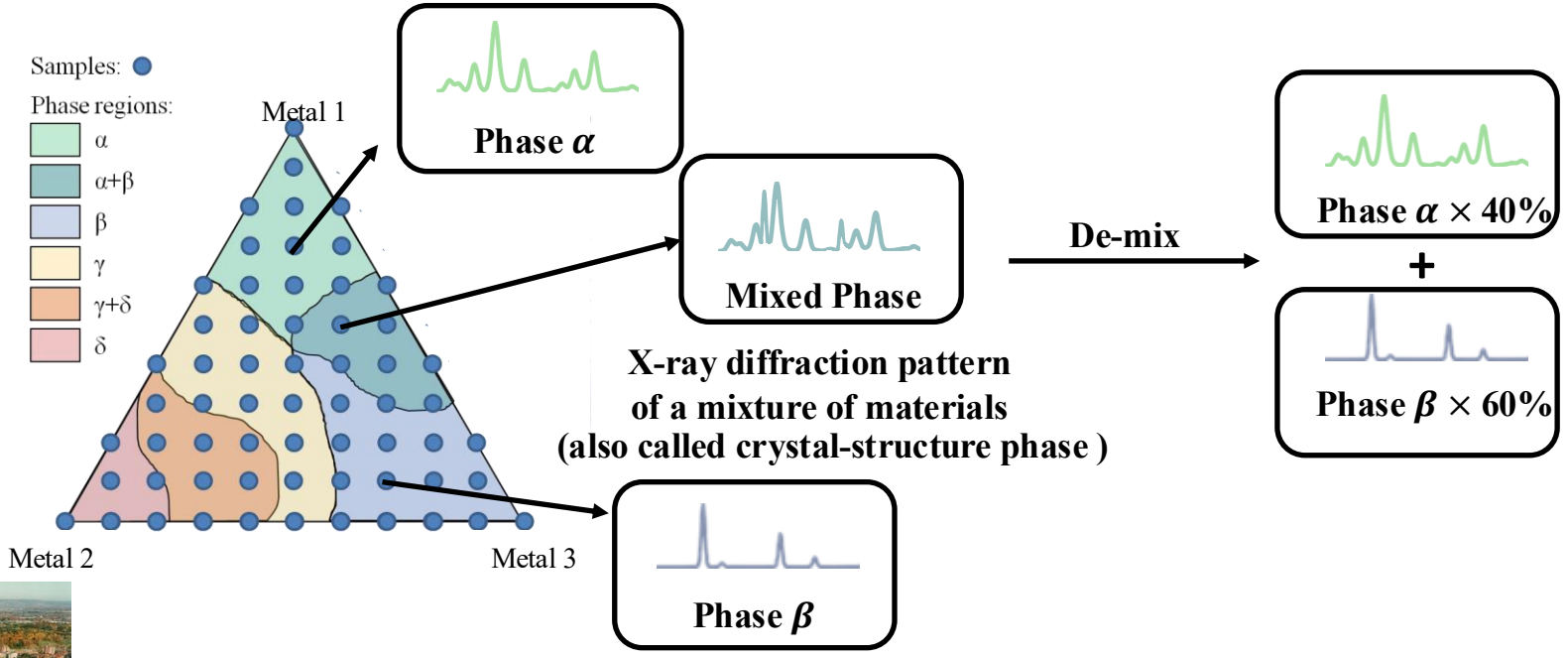
***m* phase regions**
- ***k* pure regions**
- ***m-k* mixed regions**

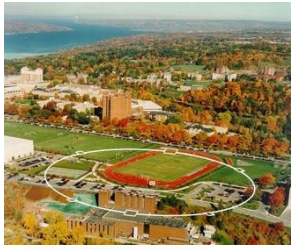
XRD pattern characterizing pure phases





Crystalline Phase Mapping Problem: Unsupervised Pattern De-Mixing





Prior Scientific Knowledge (for constraining the system):

- **Pure phase prototypes** { phase region connectivity
- **Thermodynamic rules** { A mixture contains at most 3 pure phases
  …

CHESS

Cornell University
Cornell High Energy Synchrotron Source

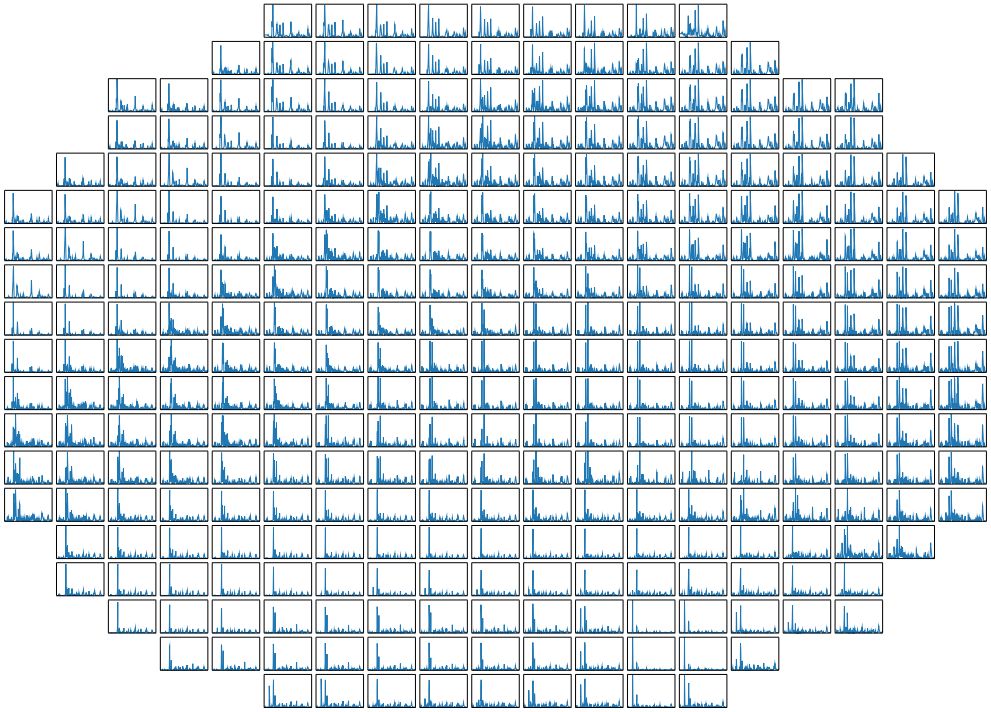
Crystal-Structure Phase Mapping:

Core longstanding problem in Materials Science: inferring crystal structures from X-ray diffraction patterns

307 Unlabeled X-Ray Diffraction Patterns of Bi-Cu-V Oxide System

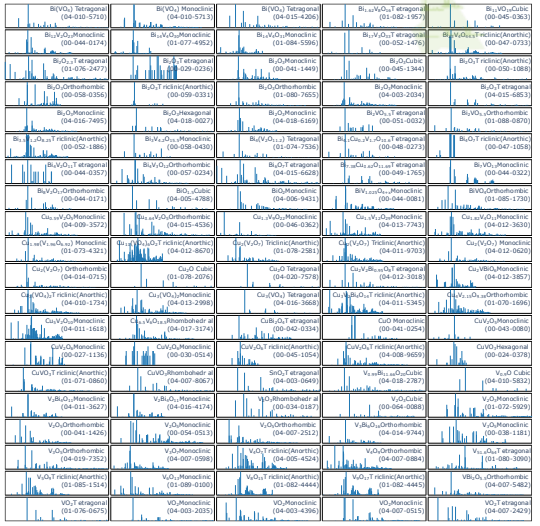
(beyond human expert capabilities, previously unsolved)

Deposition space



Prior knowledge

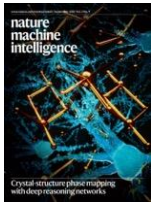
"Stick patterns" (Ideal noise free patterns) of known crystal phases from other systems involving Bi, Cu, or V:



Thermodynamic rules:

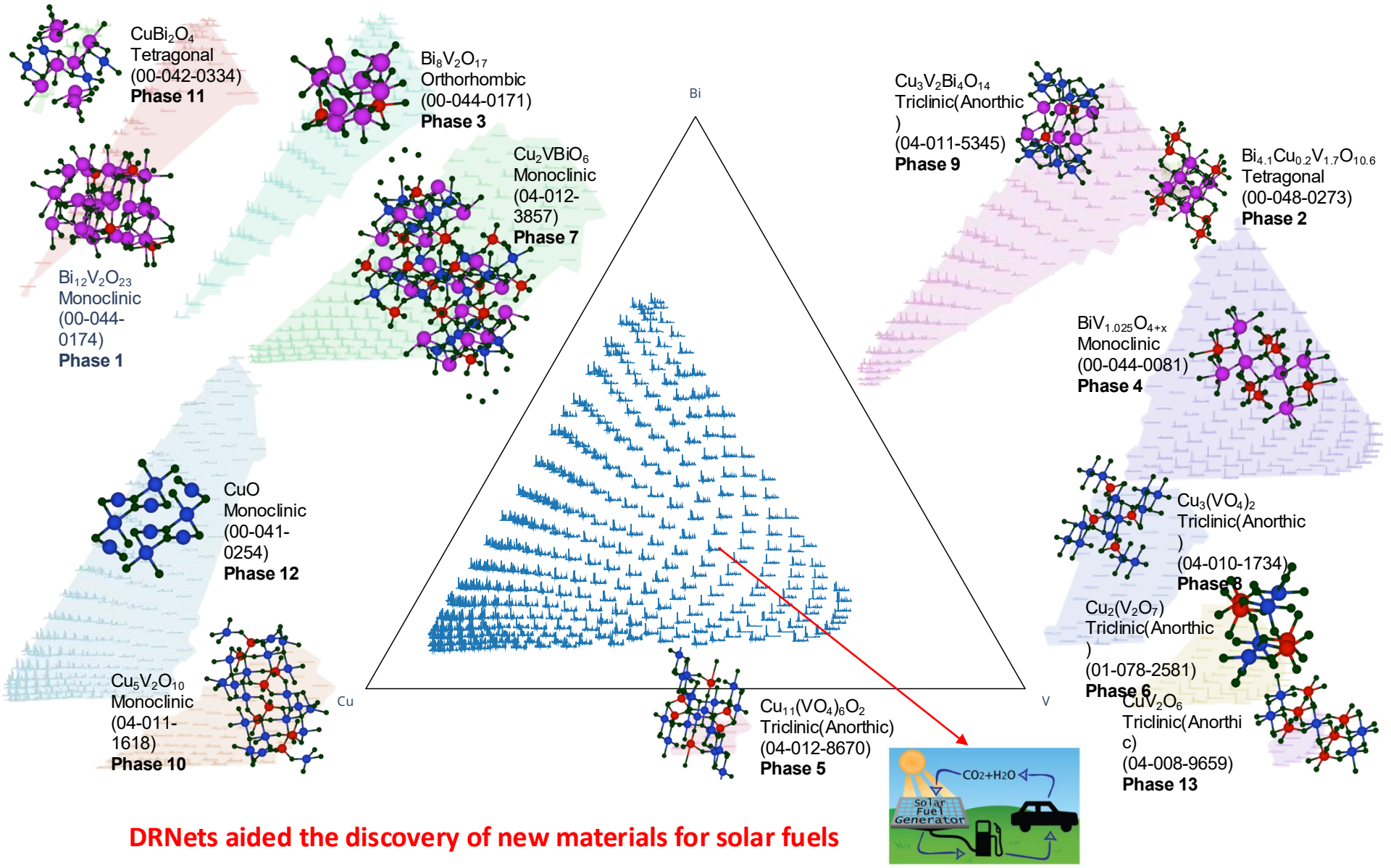
E.g., Gibbs' rules, Gibbs' alloy rule, phase connectivity.

(Unsupervised task: no training data)



Di Chen, Yiwei Bai, Sebastian Ament, Wenting Zhao, Dan Guevarra, Lan Zhou, Bart Selman, R. Bruce van Dover, John M. Gregoire, Carla P. Gomes. *Nature Machine Intelligence, September 2021. [pre-print in arXiv].*

Deep Reasoning Nets solved the previously unsolved Bi-Cu-V oxide system
13 crystal structures (pure phases) (pure phases) and 19 phase regions

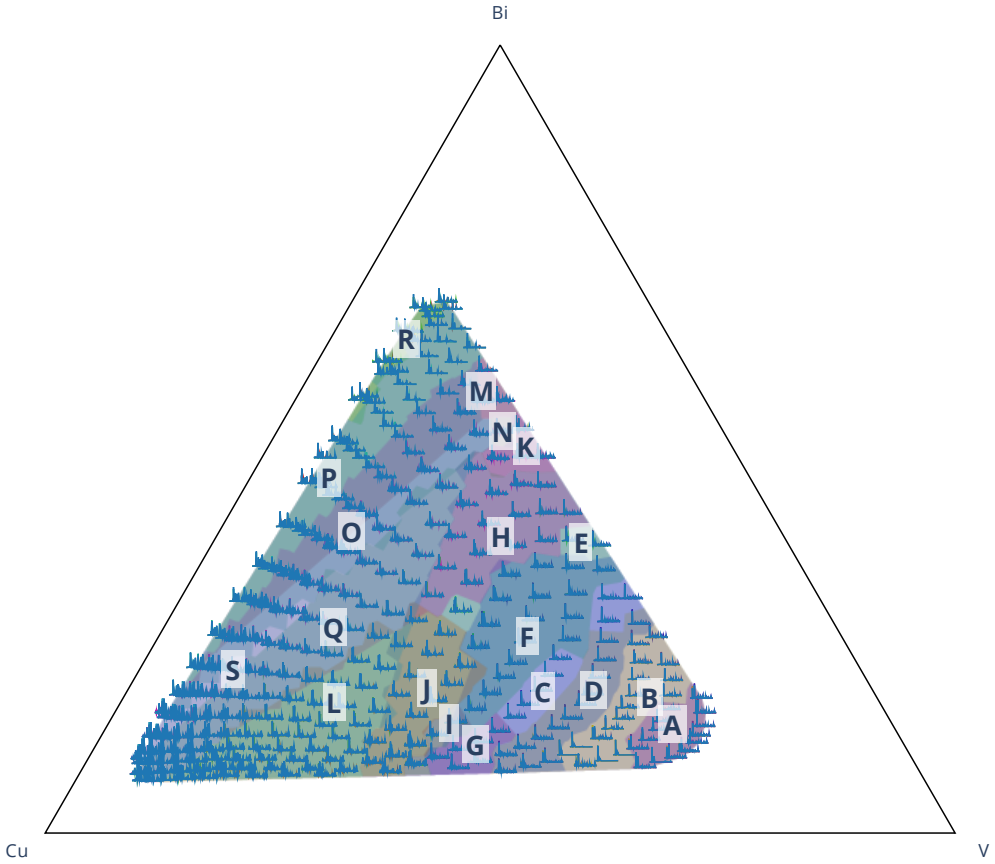


DRNets aided the discovery of new materials for solar fuels

Credit: John Gregoire

Deep Reasoning Nets solved the previously unsolved Bi-Cu-V oxide system

by combining deep learning with reasoning about known phases and thermodynamic rules



Region	Phases
A	4,6,13
B	4,6
C	4,8
D	4,6,8
E	4,7
F	4,7,8
G	4,5,8
H	4,7,9
I	4,5,7
J	4,7,10
K	2,4,9
L	7,10,12
M	1,3,9
N	2,3,9
O	3,9,12
P	1,3,12
Q	7,9,12
R	1,3,11
S	9,12

19 phase fields (phase mixtures)

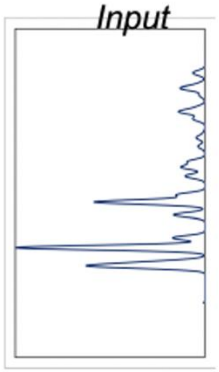


PSI-DRNet

Physics and Symmetry Informed Deep Reasoning Net: Crystal-Structure Phase Mapping (Unsupervised)

X-Ray Diffraction

Input

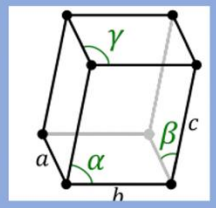


Encoder
GNN

Inverse model



Interpretable Latent Space



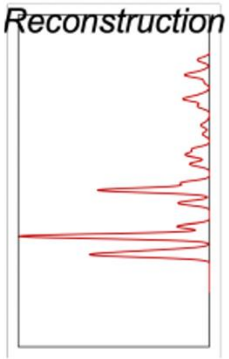
Crystal-structure embeddings:
Length of the edges: a,b,c
Angle between edges: α, β, γ

Reasoning Modulo (Thermodynamic Rules)

Decoder
Differentiable Bragg's Law Reasoning + Gaussian Mixed Model

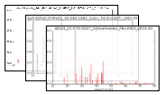
Forward Model

Reconstruction



Reconstruction Loss + Reasoning Loss

Prototypes (ICDD stick patterns)



&

Assumption of underlying GMMs for the experimental XRD patterns.

Thermodynamics Rules:

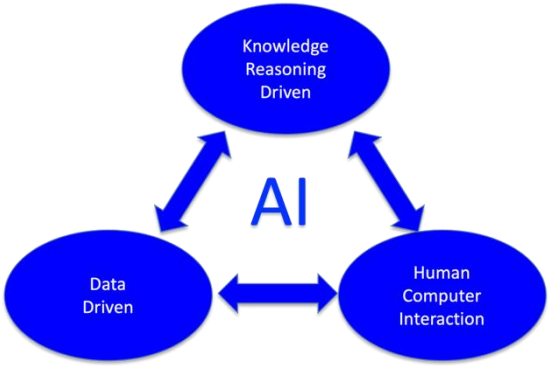
- Gibbs Phase Rule
- Gibbs-Alloying Rule
- Phase Field Connectivity
- Multiplicative Shifting

Prior Knowledge

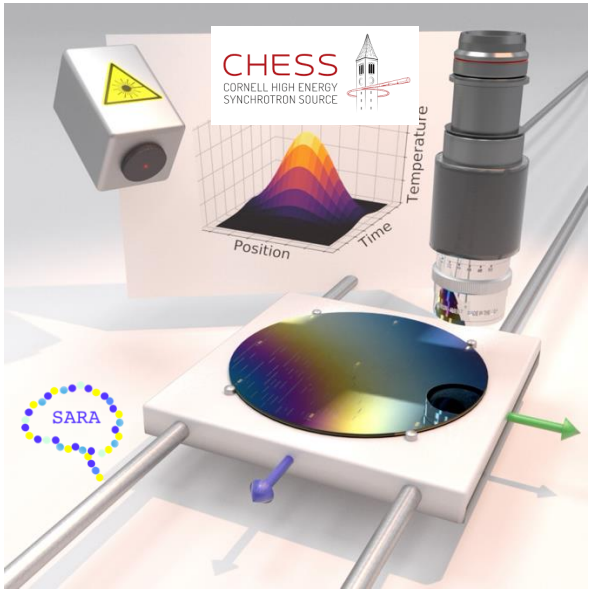
Big Picture: Knowledge-centric multi-agentic framework

Scientific Autonomous Reasoning Agent

Integrating Materials Theory, Experiment and AI

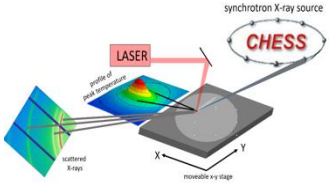






AI Robotic Scientist

Formulation of Hypotheses,
Planning, Execution and
Interpretation of Experiments
with Production of New Knowledge



Depiction of SARA controlling
laser spike annealing synthesis experiments
and the collection and interpretation of optical and XR
data

DRNets and PSI-DRNet, solved previously unsolved chemical systems,
enabling the discovery of
fuel cell and solar fuel materials and other materials

MURI

Cornell University

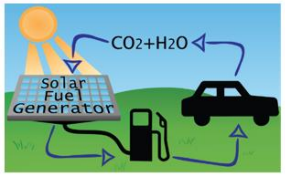
Northwestern University

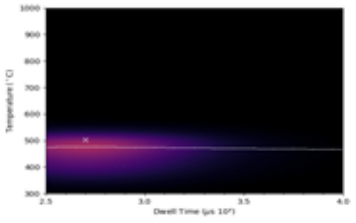
University of Colorado Boulder

Caltech

#4 on Top-10 "Coolest Science, Technology Advances" (2020)

AAAI, IJCAI, ICML, Neurips, …





Innovative AI Award

SARA Team



Cornell

Sebastian Ament (CS)
Maximillian Amsler (MSE)
Rich Bernstein (CS)
Yiwei Bai (CS)
Di Chen (CS)
Yimeng Min (CS)
Ming-Chiang Chang (MSE)
Duncan Sutherland (MSE)
Katie Gann (MSE)
Aine Connolly (MSE)
Cameron Gorsak (MSE)
Prof. Michael Thompson (MSE)
Prof. R. Bruce van Dover (MSE)
Prof. Carla Gomes (CS)

Caltech

Dr. Lan Zhou (MSE)
Dan Gueverra (MSE)
Dr. John Gregoire (MSE (now @ Lila Sciences)

CHESS

Dr. Arthur Woll (MSE)
Dr. Louisa Smieska (MSE) (not in pic)

Northwestern Univ.

Prof. Chris Wolverton MSE) (not in pic)

Uni. Colorado Boulder

Prof. Alex Zunger MSE) (not in pic)

MURI

Cornell University

Northwestern University

University of Colorado Boulder

Caltech

Schmidt Sciences

AFRL THE AIR FORCE RESEARCH LABORATORY LEAD | DISCOVER | DEVELOP | DELIVER

FNSNF FONDS NATIONAL SUISSE DE LA RECHERCHE SCIENTIFIQUE

NSF

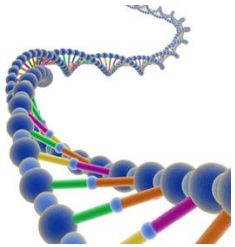
CORNELL CENTER FOR MATERIALS RESEARCH

CNF Cornell NanoScale Science and Technology Facility

CHESS CORNELL HIGH ENERGY SYNCHROTRON SOURCE

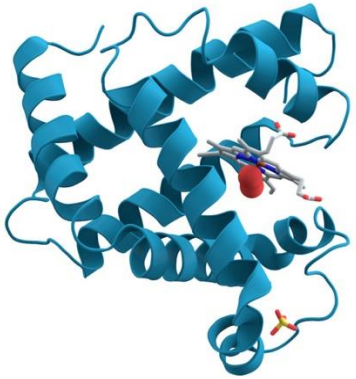
Cornell University Center for Advanced Computing

Unraveling the “dark matter” of metabolomics using mass-spectrometry

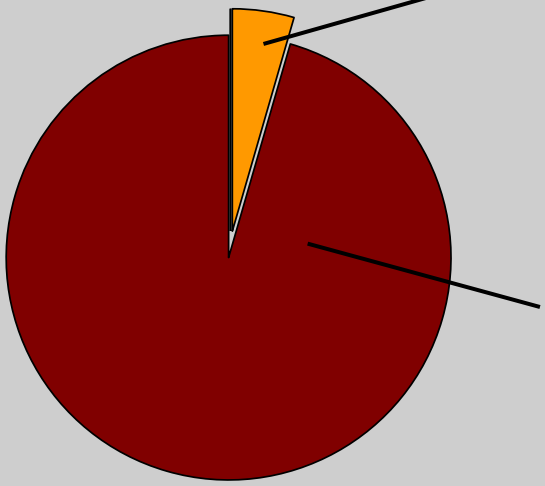


20,000 genes

The Chemistry of Life



More than 20,000 proteins



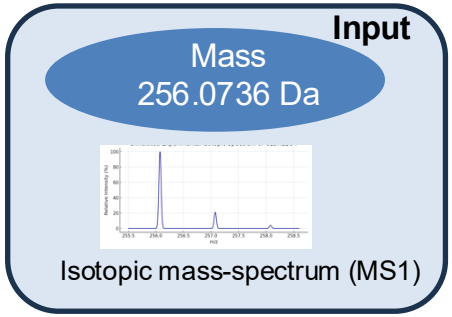
The metabolome

AI Metabolomics

Automating the rapid identification of metabolites from mass-spectrometry

We are studying three problems:

1 – Compound to formula - Predict the molecular formula from MS1 data

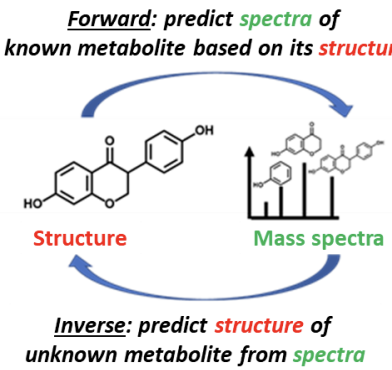


The Coin Change Problem

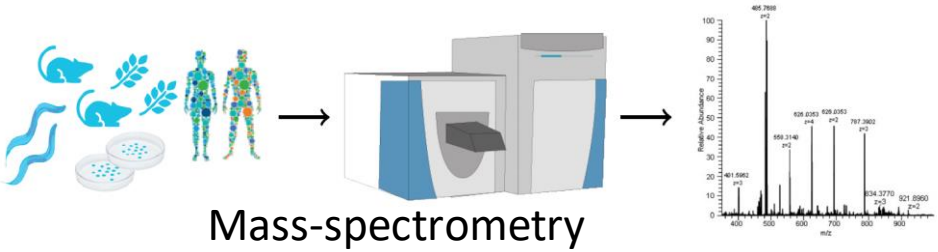
2 - Forward Problem: Given molecular structure, predict its mass-spectrum (MS2)

3 - Inverse Problem: Given mass-spectrum (MS2) predict structure

$C_{15}H_{12}O_4$



Briefly



AI2050

Schmidt Sciences

Coin Change Problem (combination variation)

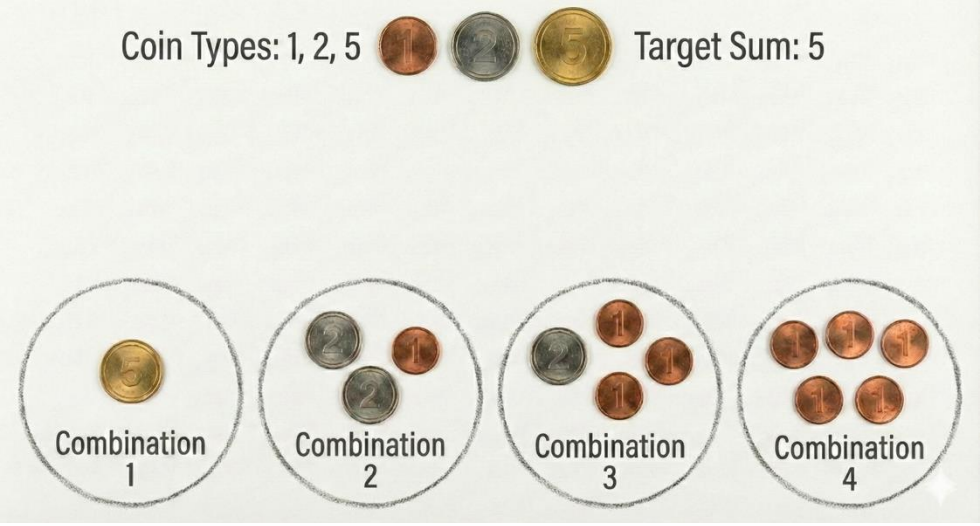
Input:

→ *A set of **n** coin types, where each type T_i has a given value v_i ;*

→ *A target sum S.*

Output:

All possible combinations (with unlimited repetitions) of the coins whose values sum exactly to the target sum S.



Input: *Molecular mass*

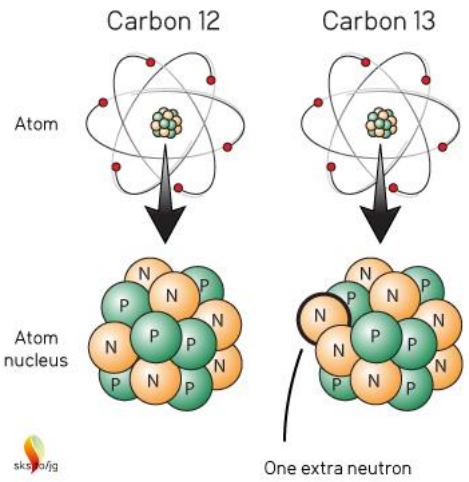
Output: Molecular formula, i.e., elements and atom counts that match the mass

$C_{15}H_{12}O_4$

Challenge: Molecular weight alone is insufficient to identify a molecule

*Potentially **thousands of molecular formulae** are possible for **one molecular weight** (even with 4 digits precision)*

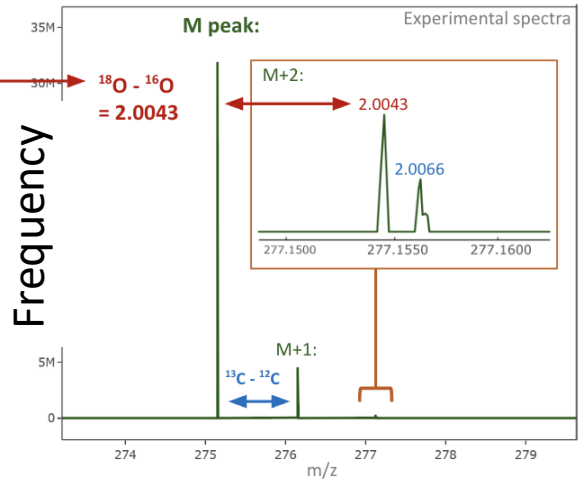
Isotopic patterns aid molecular formula determination



Element	Isotope	Mass	Abundance
H	^{1}H	1.007825	0.999885
H	^{2}H	2.014102	0.000115
C	^{12}C	12.000000	0.989300
C	^{13}C	13.003355	0.010700
N	^{14}N	14.003074	0.996360
N	^{15}N	15.000109	0.003640
O	^{16}O	15.994915	0.997570
O	^{17}O	16.999132	0.000380
O	^{18}O	17.999160	0.00205(
F	^{19}F	18.998403	1.00000(
P	^{31}P	30.973761	1.00000(
S	^{32}S	31.972071	0.94990(
S	^{33}S	32.971459	0.00750(
S	^{34}S	33.967867	0.04250(
S	^{35}S	35.000000	0.00000(
S	^{36}S	35.967081	0.00010(
Cl	^{35}Cl	34.968853	0.75760(
Cl	^{36}Cl	36.000000	0.000000
Cl	^{37}Cl	36.965903	0.242400

Monoisotopic peak:
Peak at which every element is its most abundant isotope

^{12}C - ^{13}C = 1.0033 → M+1 – M = 1.0033



Carbon isotopes

Isotopic patterns provide fingerprints of compounds showing up on mass spectra (when using ultra high-resolution mass spectrometer)

ThermoFisher SCIENTIFIC

Abstraction: Probabilistic Coin Change Problem

Input:

→ *A set of **n** coin types **T**, where each type **T_i** has an associated **discrete weight distribution Wi***

→ *A **noisy empirical distribution over the total sum S.***

Output: *The (latent) coin combination that best explains the observed distribution.*



Challenges in Inferring Molecular Formula from Mass Spectrometry

- *Experimental noise brings stochastic variations to measured isotopic patterns. Learning from data is required.*
- *No labeled dataset available. Expert labeling is time consuming.*
- *Adherence to chemistry principles: Need to reasoning about chemistry and combinatorics*

Unsupervised learning grounded in scientific knowledge.

Forward / Inverse sub-problems for molecular formula determination

Forward Problem: Assume you know the formula → Isotopic pattern

Isotope patterns (theoretical intensities of the peaks) follow the multinomial distribution:

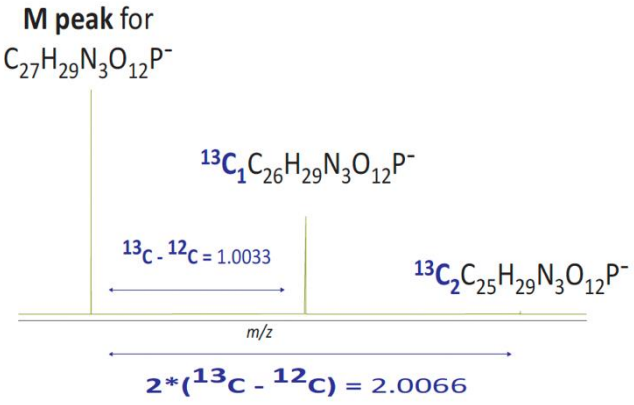
$$\mathbb{P}(s \mid \mathbf{z}) = \prod_{i=1}^{n} \frac{e_i!}{\prod_{j=1}^{m_i} c_{i,j}!} \prod_{j=1}^{m_i} p_{i,j}^{c_{i,j}} = \prod_{i=1}^{n} \prod_{j=1}^{m_i} \frac{e_i! p_{i,j}^{c_{i,j}}}{c_{i,j}!}$$

- e_i: the number of atoms of element i in the formula.
- $c_{i,j}$: the number of atoms as isotope j for element i.
- $p_{i,j}$: the abundance of isotope j for element i.

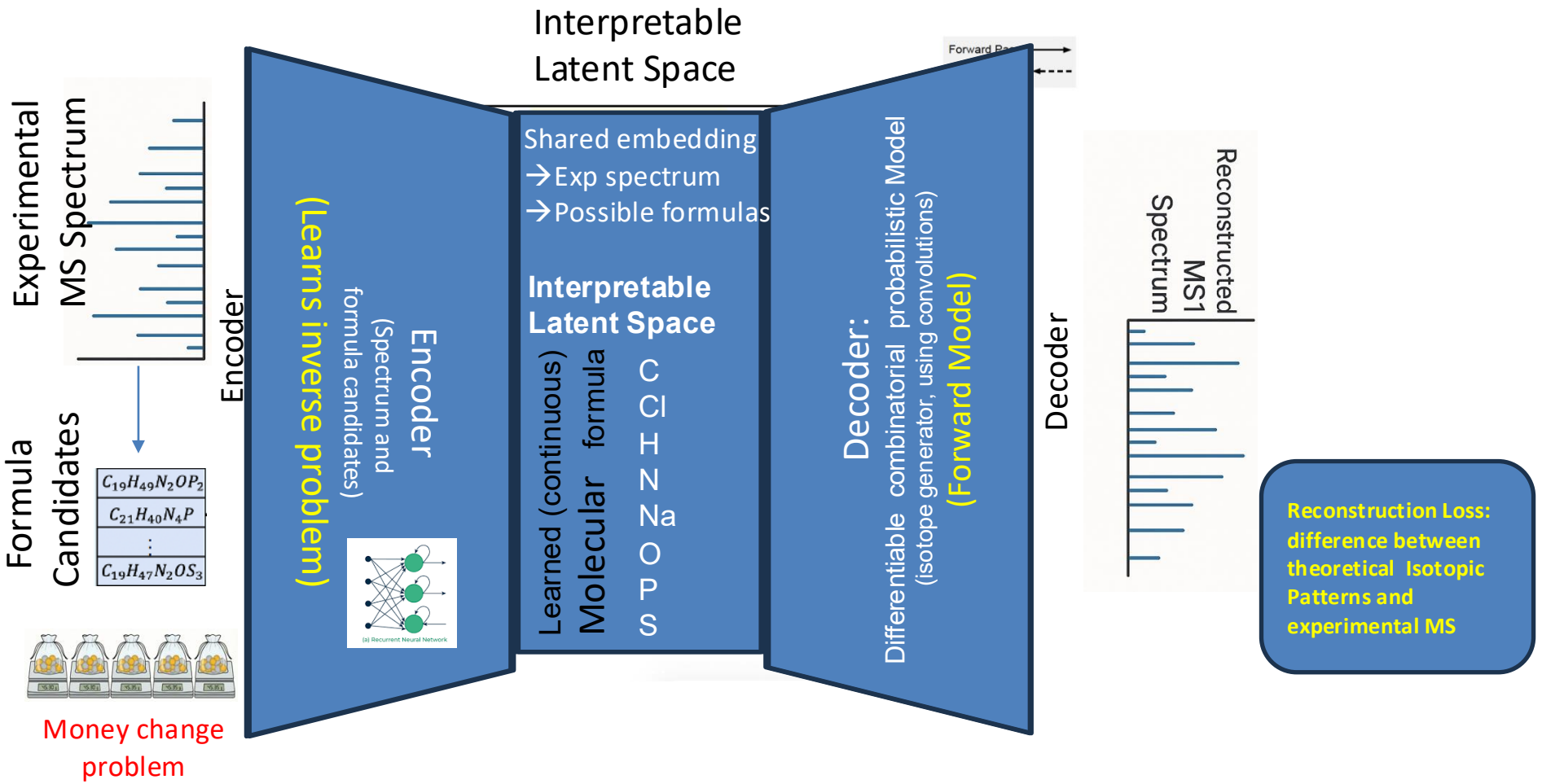
Inverse Problem (Goal): Assume you know the experimental Isotopic Pattern → formula

Infer the molecular formula that has produced the isotopic pattern as measured on the

Experimental mass spectrum.



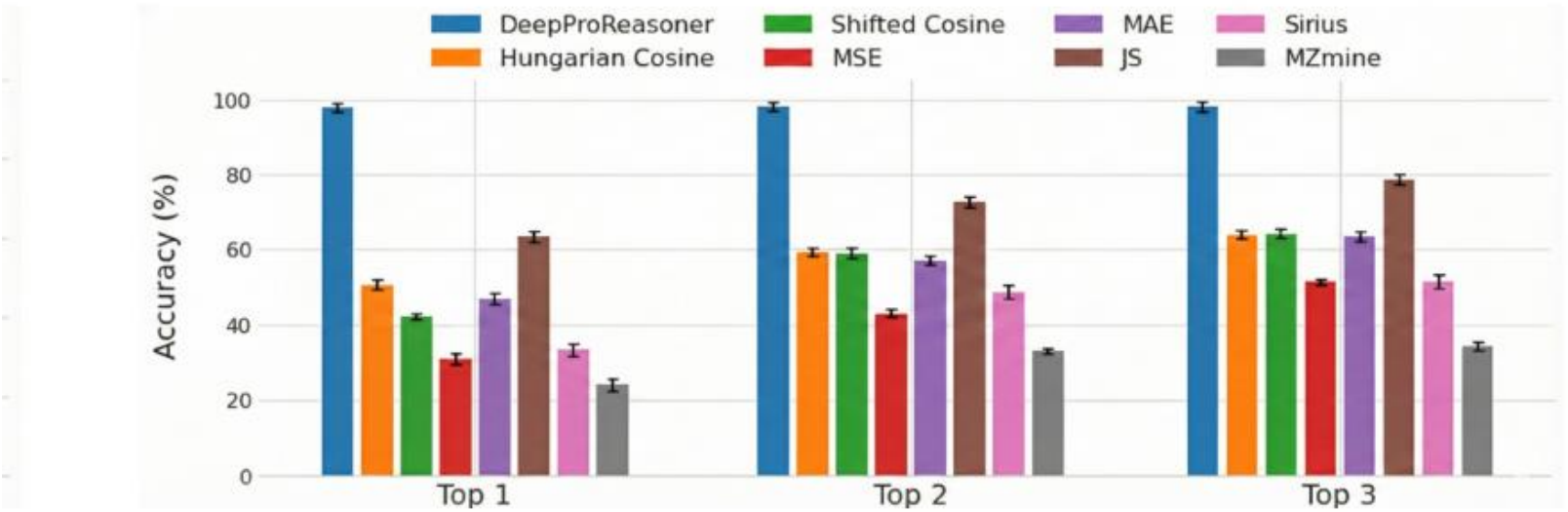
DeepProReasoner: Unsupervised Probabilistic Coin Change Problem Molecular Formula Inference from Mass Spectra



Zhongdi Qu, Yingheng Wang, Utku Acikalin, Aaron M. Ferber, Goncalo J. Gouveia, Brandon Bills, Guohui Li, Joshua Kline, Sunandini Yedla, Xiao Wang, Frank C. Schroeder, Carla P. Gomes *AAAI 2026*

Manuscript in preparation for submission to Nature Methods.

Results on real data outperform academic and commercial State of the Art



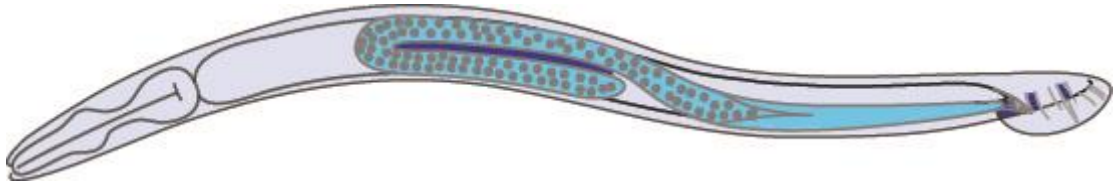
Top K (*mouse liver)*

DeepProReasoner

DeepProReasoner outperforms all other academic and commercial methods

Zhongdi Qu, Yingheng Wang, Utku Acikalin, Aaron M. Ferber, Goncalo J. Gouveia, Brandon Bills, Guohui Li, Joshua Kline, Sunandini Yedla, Xiao Wang, Frank C. Schroeder, Carla P. Gomes ***AAAI 2026***

Manuscript in preparation for submission to ***Nature Methods*****.**



TEAM MEMBERS

Schroeder Lab

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Dr. Goncalo Gouveia
Dr. Tyler J. Schwertfeger (Shmidt AI 4 Sci. Postdoc)
Dr. Andreas H. Ludewig (now in Tel Aviv)
Dr. Tae Hyung Won (Cha Univ., Korea)
Dr. Alexander B. Artyukhin
Dr. Russell N. Burkhardt
Dr. Henry H. Le
Dr. Chester J. Wrobel
Dr. Yan Yu
Dr. Jingfang Yu
Dr. Brian J. Curtis
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Gomes Lab

Utku Acikalin
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Yingheng Wang
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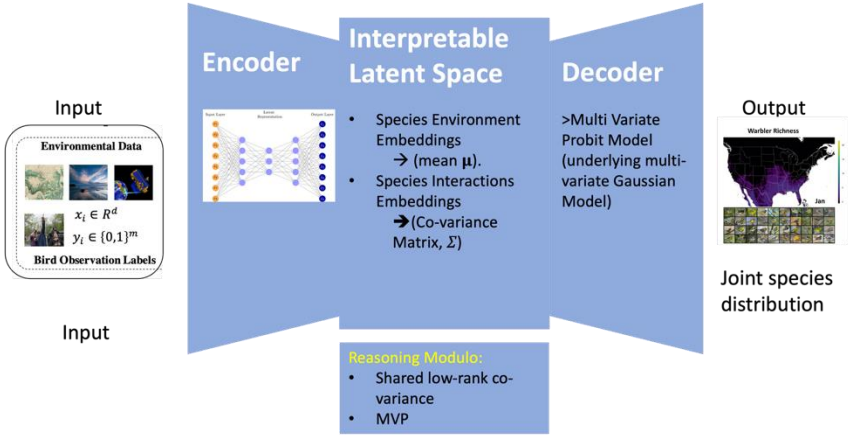
Carla P. Gomes

CORNELL UNIVERSITY FOUNDED A.D. 1865

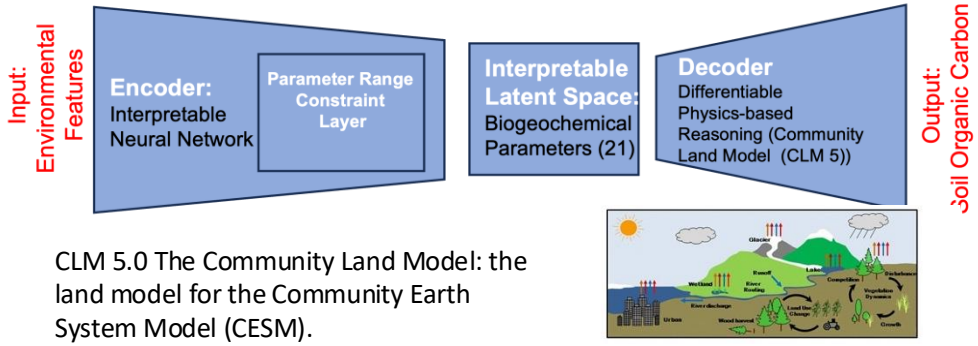
BTI

Prediction Tasks

Joint Species Distributions (Supervised)



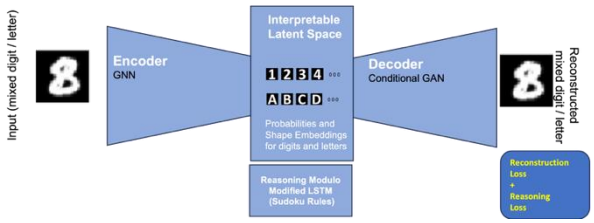
Predicting and Understanding the Soil Organic Carbon Cycle (weakly supervised)



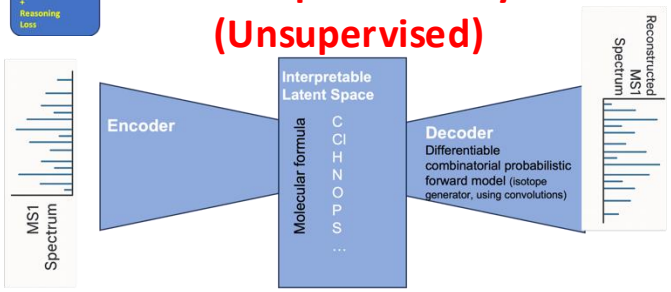
CLM 5.0 The Community Land Model: the land model for the Community Earth System Model (CESM).

Inverse Tasks

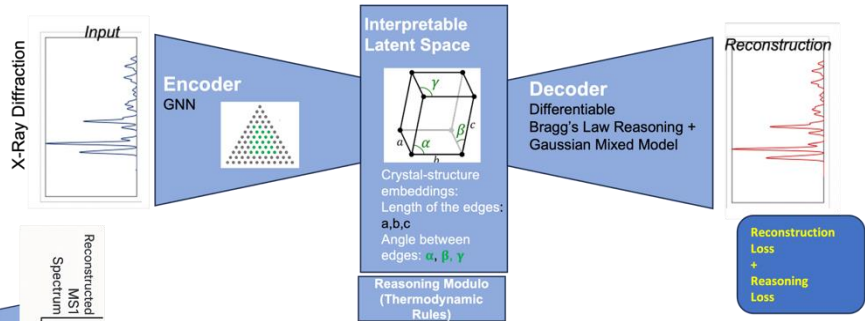
Demixing Sudokus (Unsupervised)



Inferring Molecular Formula from Mass Spectrometry (Unsupervised)

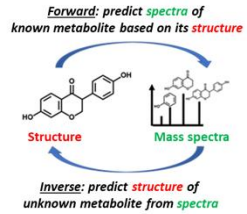


Crystal-Structure Phase Mapping (Unsupervised)



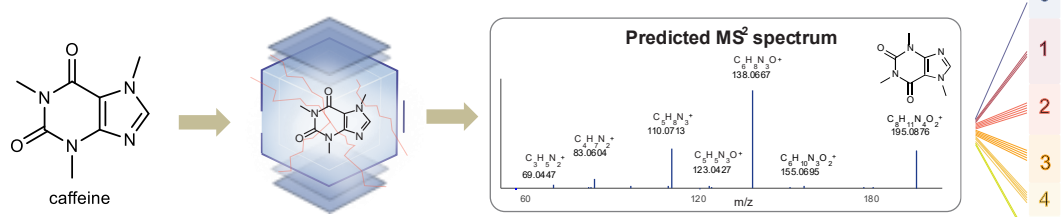
AIMe: AI Molecule Explorer Framework

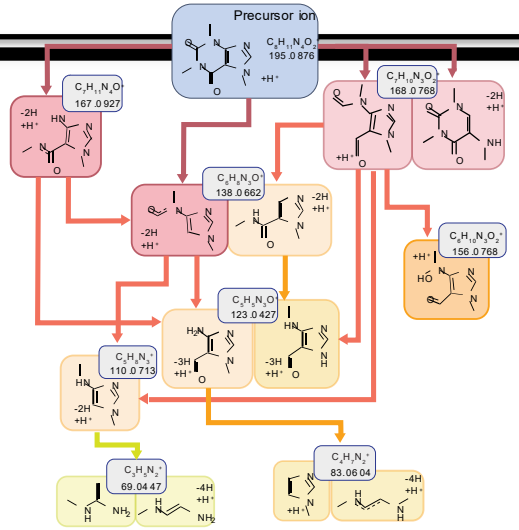
Charting the small-molecule universe from mass spectra with neuro-symbolic AI



Forward Model: Spectrum and Fragmentation Pathways Prediction

DeepMS²Reasoner:





AIMe: multi-agent neuro-symbolic AI framework for predicting, interpreting, exploring, and mapping small-molecule from MS data.

DeepMS²Reasonner

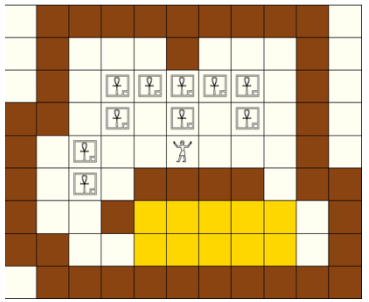
highly inspired by Sokoban

(Prototypical AI Planning problem)

AI2050

Schmidt Sciences

Sokoban

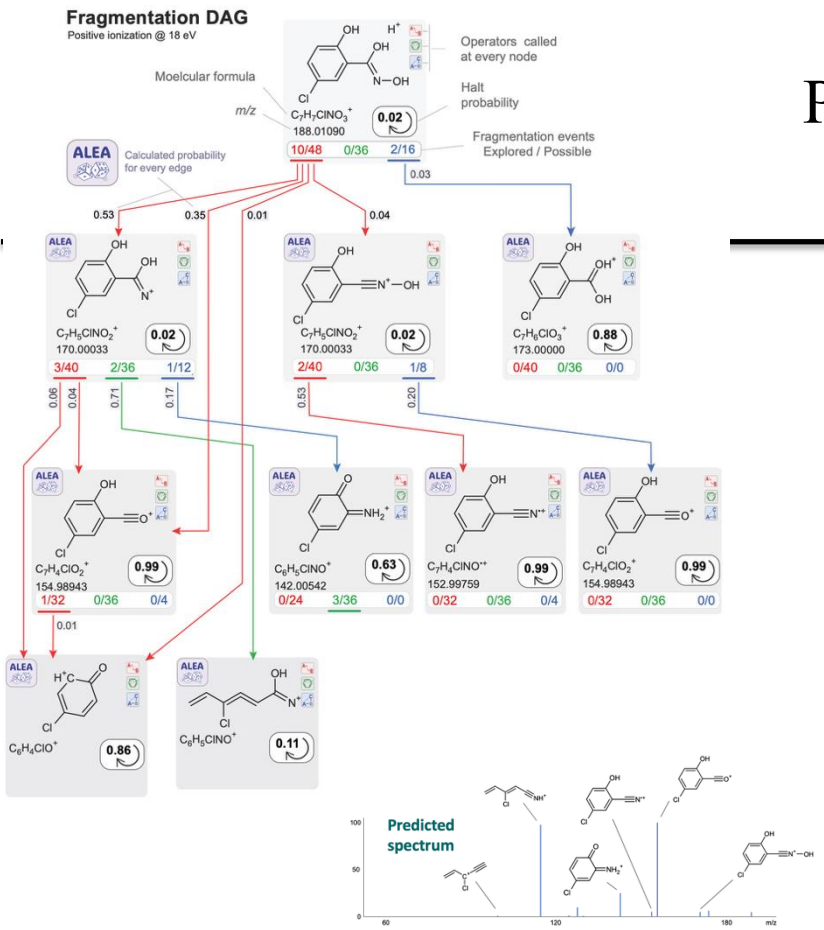


Utku U. Acikalin, Dieqiao Feng, Aaron M. Ferber, Goncalo J. Gouveia, Tyler J. Schwertfeger, Di Chen, Delia Qu, Marissa A. Fontaine, Yingheng Wang, Richard A. Bernstein, Haofan Wang, Tae-Hyung Won, Christopher N. Parkhurst, Bart Selman, David Artis, Frank C. Schroeder, Carla P. Gomes. Charting the small-molecule universe from mass spectra with neuro-symbolic AI. *bioRxiv* (2026) doi:10.64898/2026.08.05.743095.

https://www.biorxiv.org/content/10.64898/2026.08.05.743095v1

(under review)

DeepMS^{2}Reasonner: Predict Spectrum + Fragmentation DAG given molecule structure



Deep RL style

Search + Reasoning

Adaptive dynamic Direct Acyclic Graph (simulating a mass spectrometer fragmentation process)

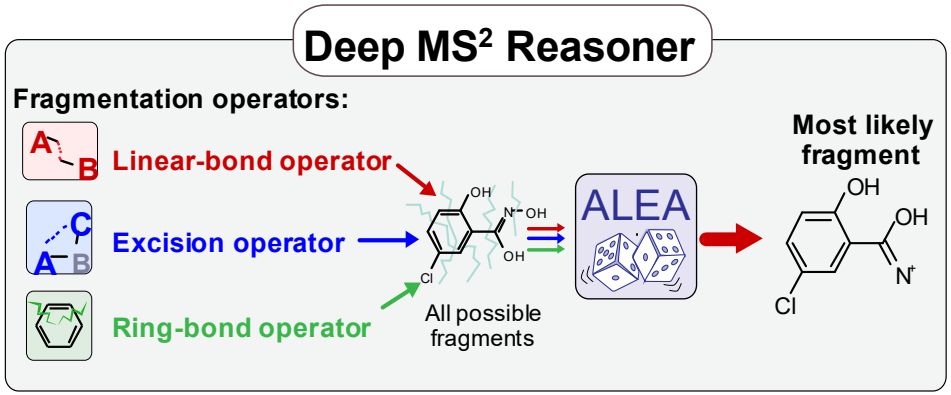
avoid exponentially many nodes

At each node

- Prune edges with too small probability
- Enforce chemistry rules
- Generate children (possible actions)

Recursive thinking: Use Markovian property for computing Mass Spectrum

Curriculum learning + randomization



Reasoning + Learning

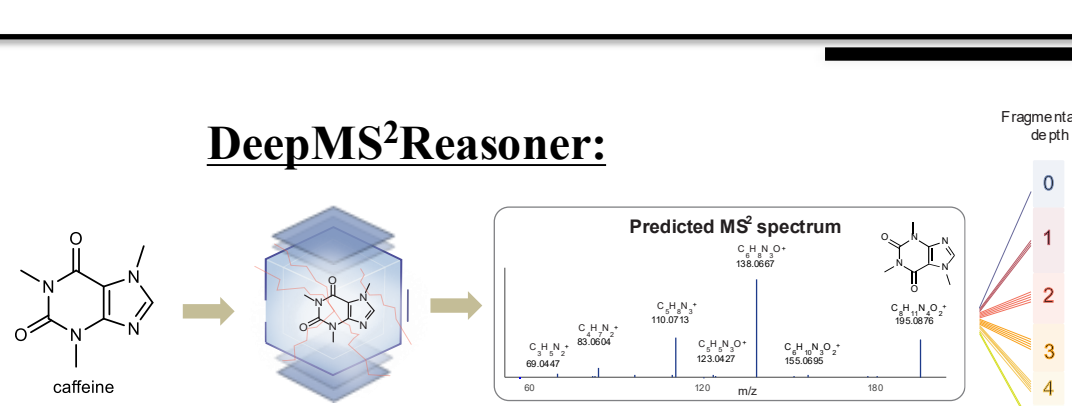
Use GraphFormer + MLP to predict fragment probabilities

AIMe: AI Molecule Explorer Framework

Charting the small-molecule universe from mass spectra with neuro-symbolic AI

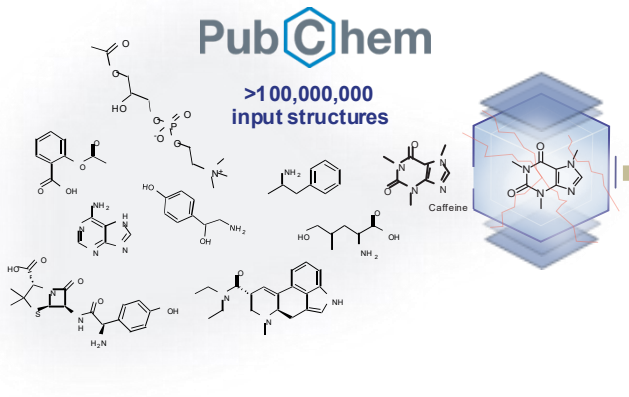
Forward Model: Spectrum and Fragmentation Pathways Prediction

DeepMS²Reasoner:



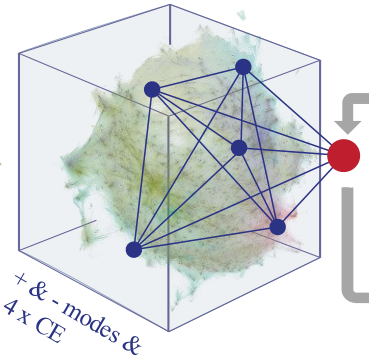
Candidate retrieval

MS²KOSMOSgenerator:

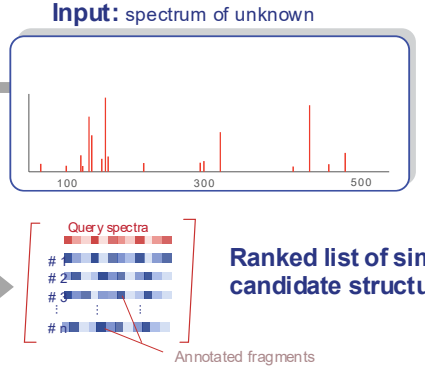


MS²KOSMOS

>800,000,000 spectra



MS²KOSMOSmapper:



Ranked list of similar candidate structures

Led to discovery of new molecular structures, confirmed by synthesis

Utku U. Acikalin, Dieqiao Feng, Aaron M. Ferber, Goncalo J. Gouveia, Tyler J. Schwertfeger, Di Chen, Delia Qu, Marissa A. Fontaine, Yingheng Wang, Richard A. Bernstein, Haofan Wang, Tae-Hyung Won, Christopher N. Parkhurst, Bart Selman, David Artis, Frank C. Schroeder, Carla P. Gomes. Charting the small-molecule universe from mass spectra with neuro-symbolic AI. *bioRxiv* (2026) doi:10.64898/2026.08.05.743095.

https://www.biorxiv.org/content/10.64898/2026.08.05.743095v1

(under review)

AI2050

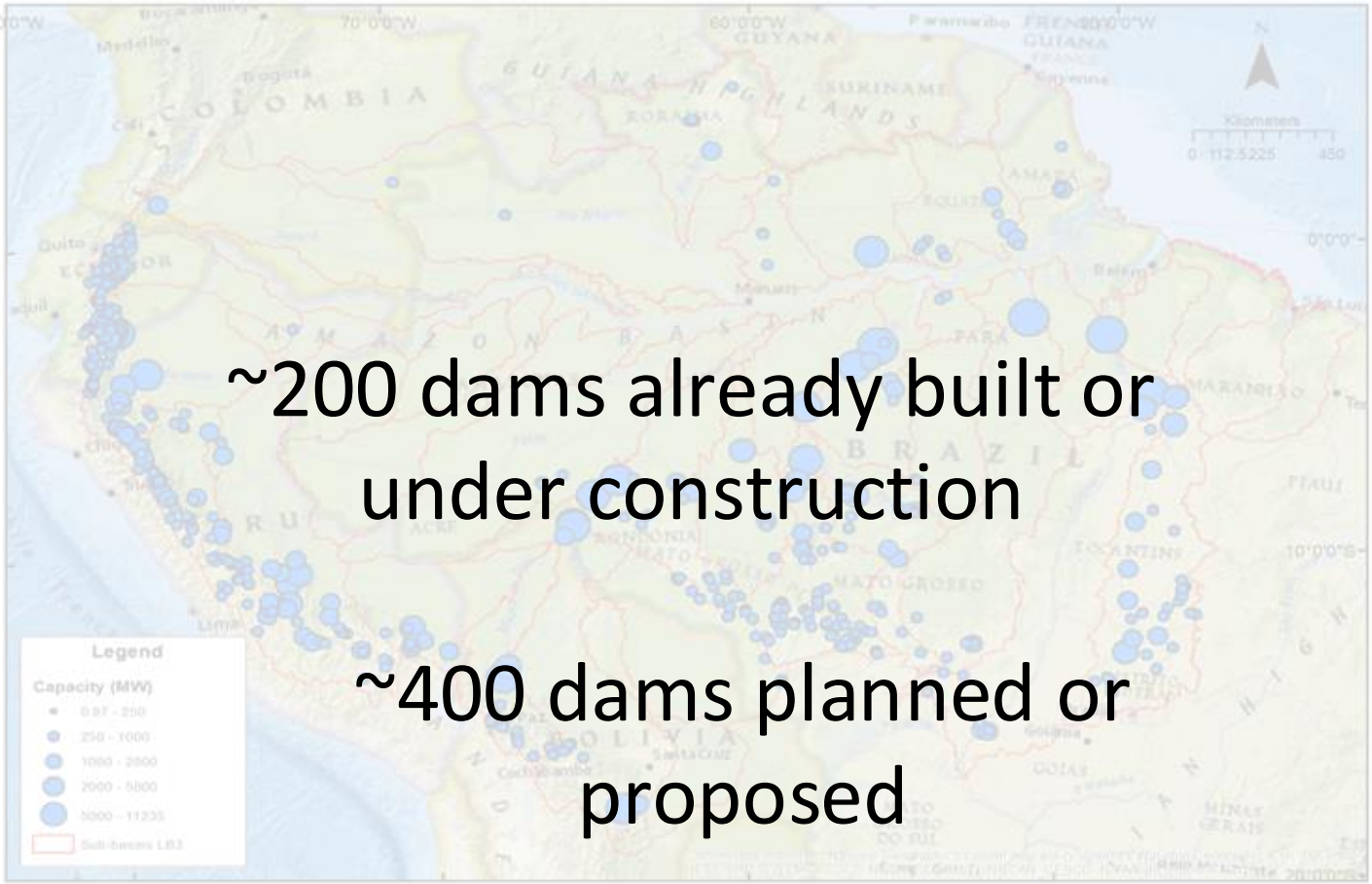
Schmidt Sciences

Decision Making



Alvaro del Campo/
The Field Museum

AI strategic planning to reduce adverse impacts of the Amazon hydropower expansion on people and nature



Almeida et al. *Nature Communications 2018* A. Flecker et al. *Science 2022.*

Almeida et al. *Nature 2022.* *CS papers CPAIOR, AAAI, IJCAI, AAMAS*

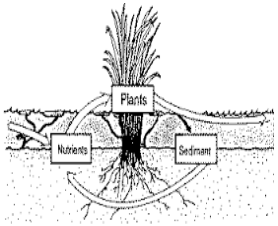
River Ecosystem Services and Impacts of Hydropower Expansion



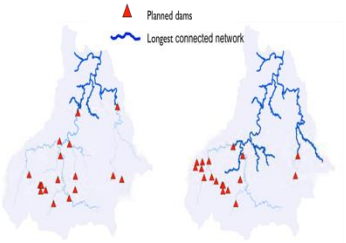
Energy



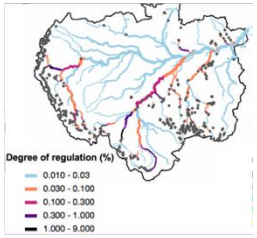
Fish Biodiversity



Sediments and Nutrients



River Connectivity (Transportation, Navigation, Fish Migration)



River Degree of Regulation



Deforestation



Displacement of populations



Flooded Farmland



GHGs

More ethical AI decision support systems:

Away from Single Objective Optimization

Multi-objective optimization and Pareto frontier:

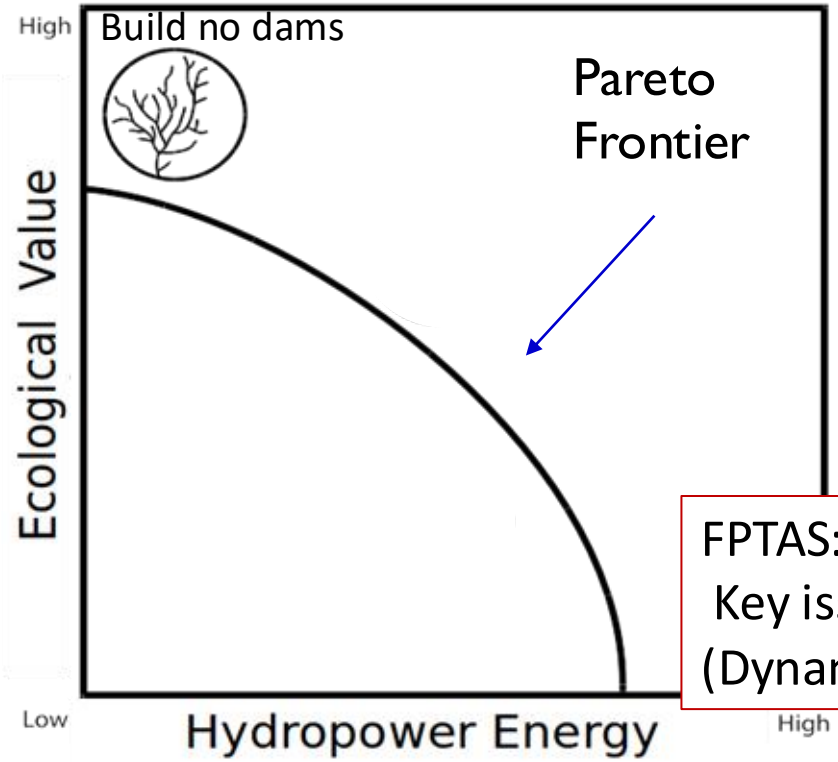
the **trade-offs** wrt to the different objectives

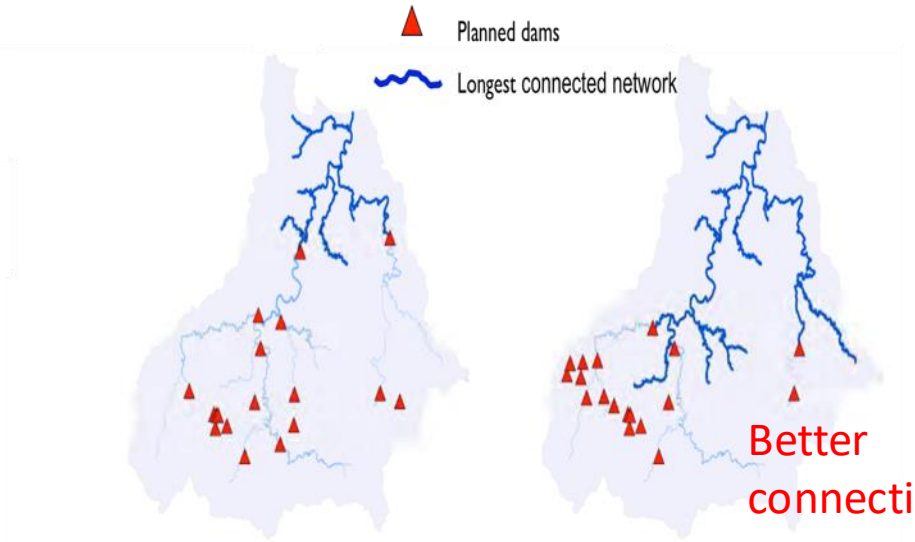
of different solutions for dam portfolios

What Hydropower Dam configurations are better than others?

Tradeoff Analysis wrt Different Criteria

Different Dam Configurations





connectivity

FPTAS: Fully Polynomial Time Approximation Scheme

Key issue: Exploit Tree Structure of Rivers

(Dynamic Programming based approach)

Our Work:

→Exact Pareto Frontier for multiple criteria

→ Efficiently Approximate the Pareto Frontier (Fully Polynomial Time Approximation Scheme: DP based approach using O(nlogn) alg. for finding non-dominated set of solutions)

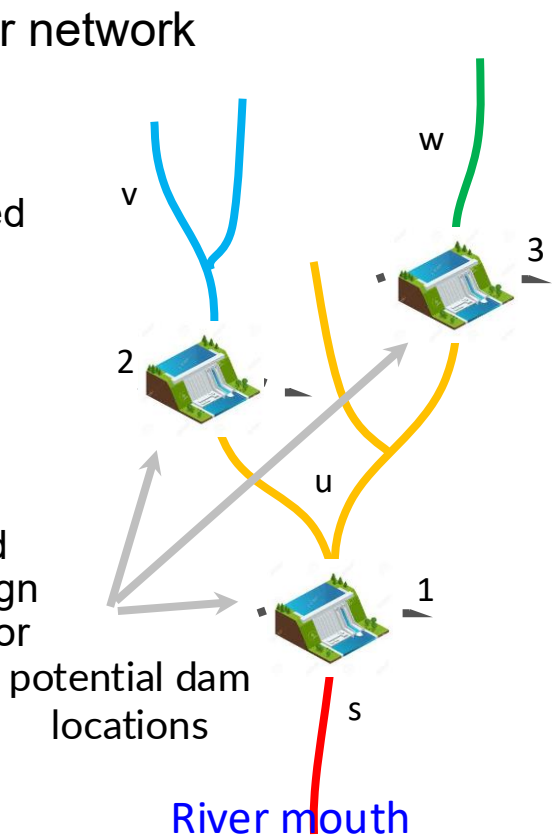
→ Hybrid strategies

Problem Representation

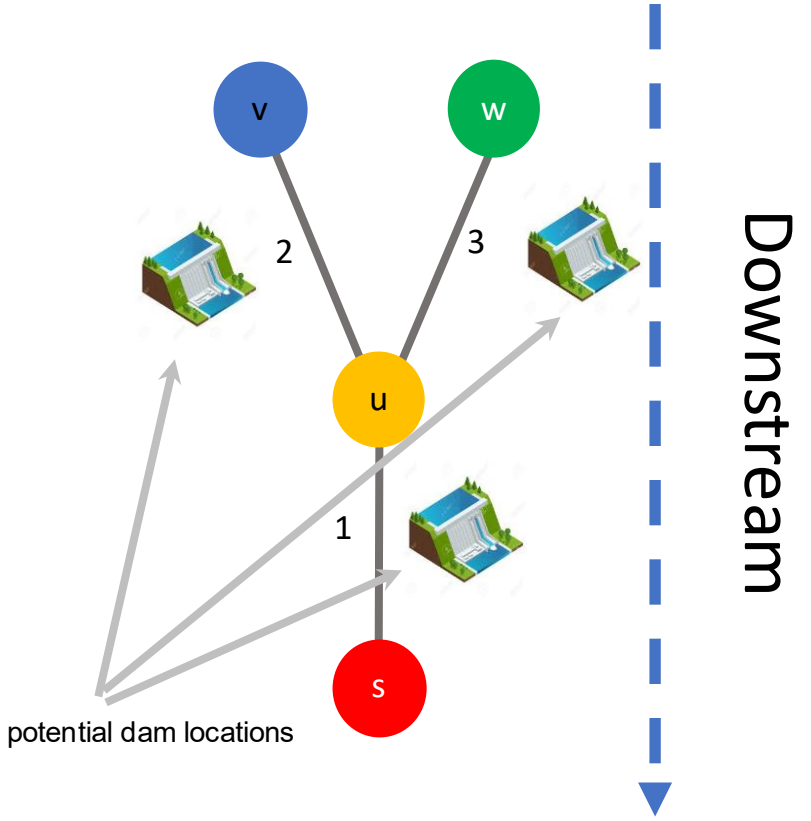
River network (left) → Rooted tree (right)

Edge – potential dam location

Node – contiguous river sub-network not affected by a potential dam (assign to the node the utilities for the different criteria)



(a). River Network

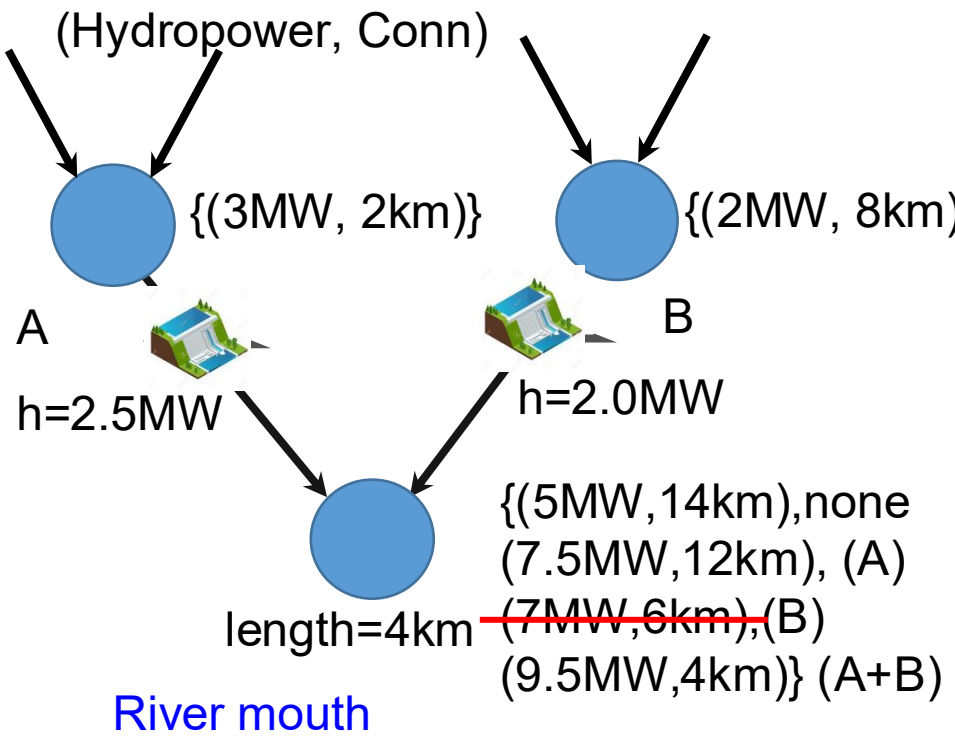


(b). Graph Representation

Wu, Gomes-Selman, Shi, Xue, García-Villacorta, Anderson, Sethi, Steinschneider, Flecker, Gomes, AAAI18

Fully Polynomial-time approximation scheme (FPTAS) Dynamic Programming Based Algorithm

- Recursively compute the Pareto-frontier from leaves to root.
- Key Insight: Only need to keep Pareto-optimal partial solutions at each node.
- Fully polynomial-time approximation scheme (FPTAS) – rounding solutions guaranteeing accuracy of (1 – ε).
- Major Bottleneck: Pruning dominated solutions.
- Pruning non-dominated solutions: divide-and-conquer $O(n(\log n)^{d-1})$



Wu, Gomes-Selman, Shi, Xue, García-Villacorta, Anderson, Sethi, Steinschneider, Flecker, Gomes, AAAI18

Gomes-Selman, Shi, Xue, Flecker, Gomes, CPAIOR18

Almeida et al. *Nature Communications 2018* A. Flecker et al. *Science 2022.*

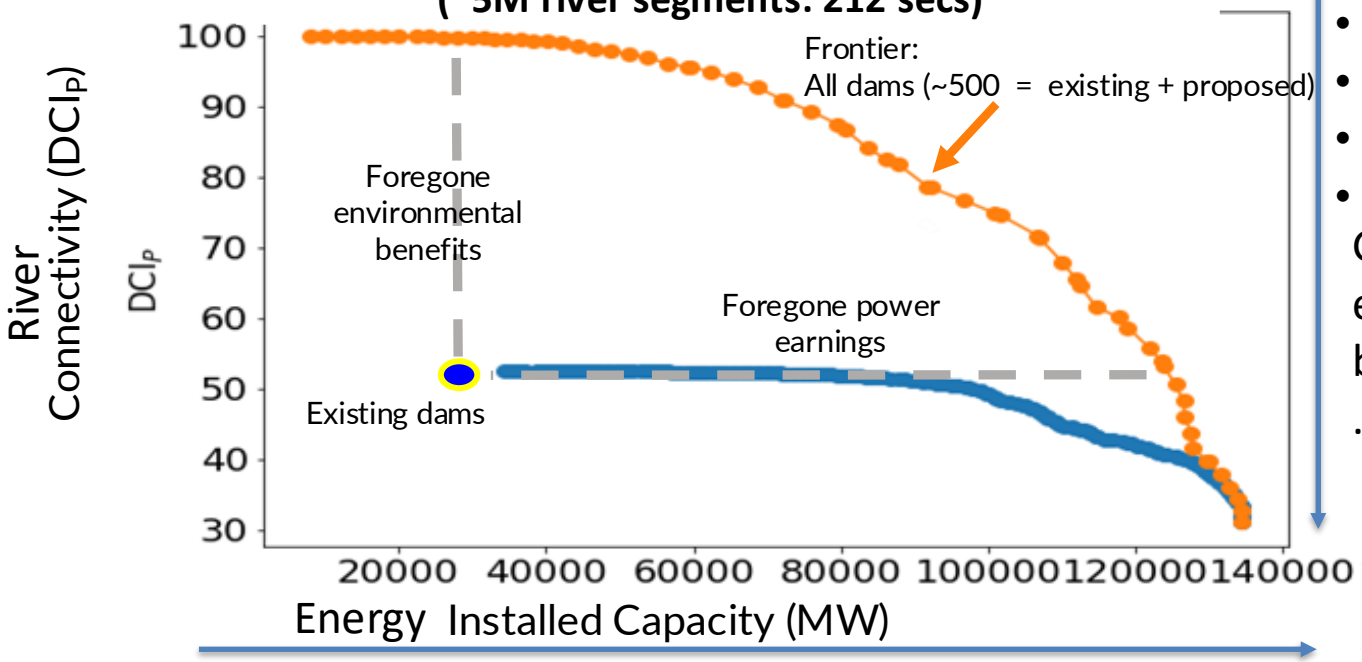
Almeida et al. *Nature 2022.* *CS papers CPAIOR, AAAI, IJCAI, AAMAS*

Message to Policy Makers: The cost of inefficient planning Connectivity

Exact Pareto Frontier Amazon Basin

(Energy vs River Connectivity)

(~5M river segments: 212 secs)



Different criteria

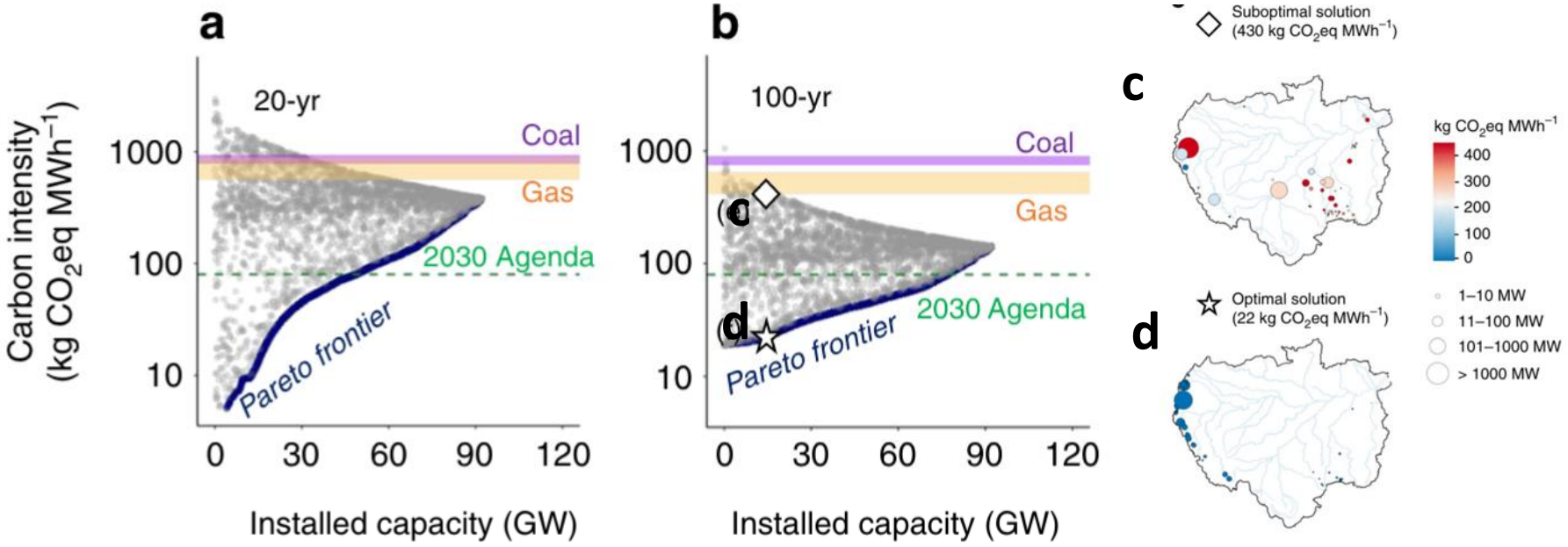
- Energy
- Connectivity
- Sediment
- Seismic risk
- Biodiversity (fish)
- Green House Gases
- Populations affected
- Cost
- …

Group **40+ collaborators**: ecologists, hydrologists, biologists, social scientists, …



Energy vs. Greenhouse Gas Emissions: Optimal Pareto frontier for proposed dams

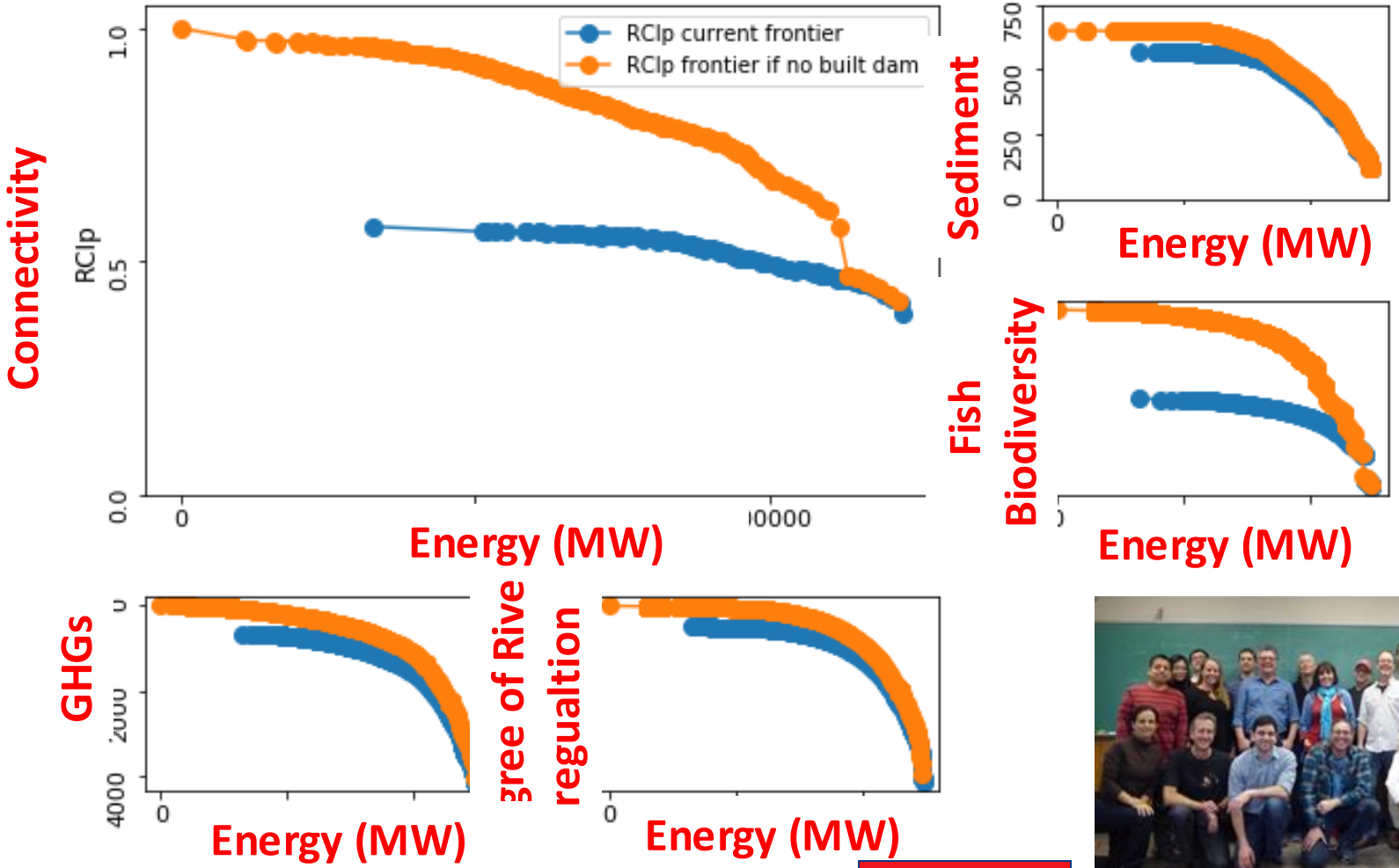
Minimization of Green Gas Emissions





Almeida, Shi, Gomes-Selman, Wu, Xue, Angarita, Barros, Forsberg, García-Villacorta, Hamilton, Melack, Montoya, PerezSethi, Gomes, Flecker, 2019

Message to Policy Makers: The cost of inefficient planning

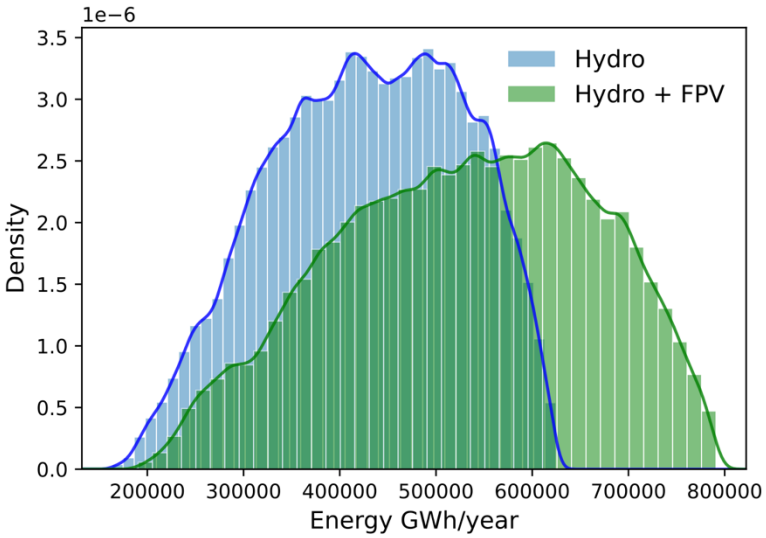




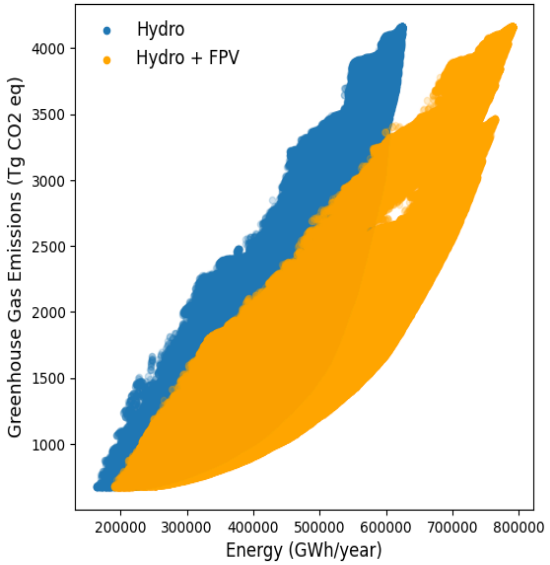
Reducing adverse impacts of Amazon hydropower expansion. in *Science, 2022*.

Science
AAAS

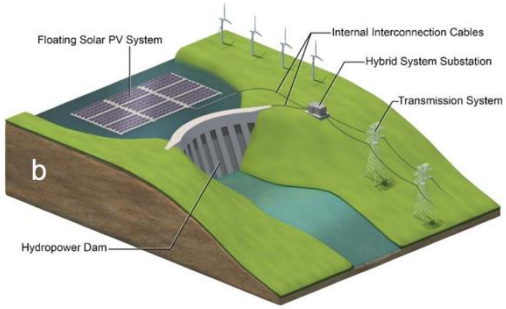
Hydropower + Floating Photovoltaics (<5% of reservoir)



Distribution of energy values among Pareto frontier solutions for hydro-power energy only versus hydro-power plus FPV



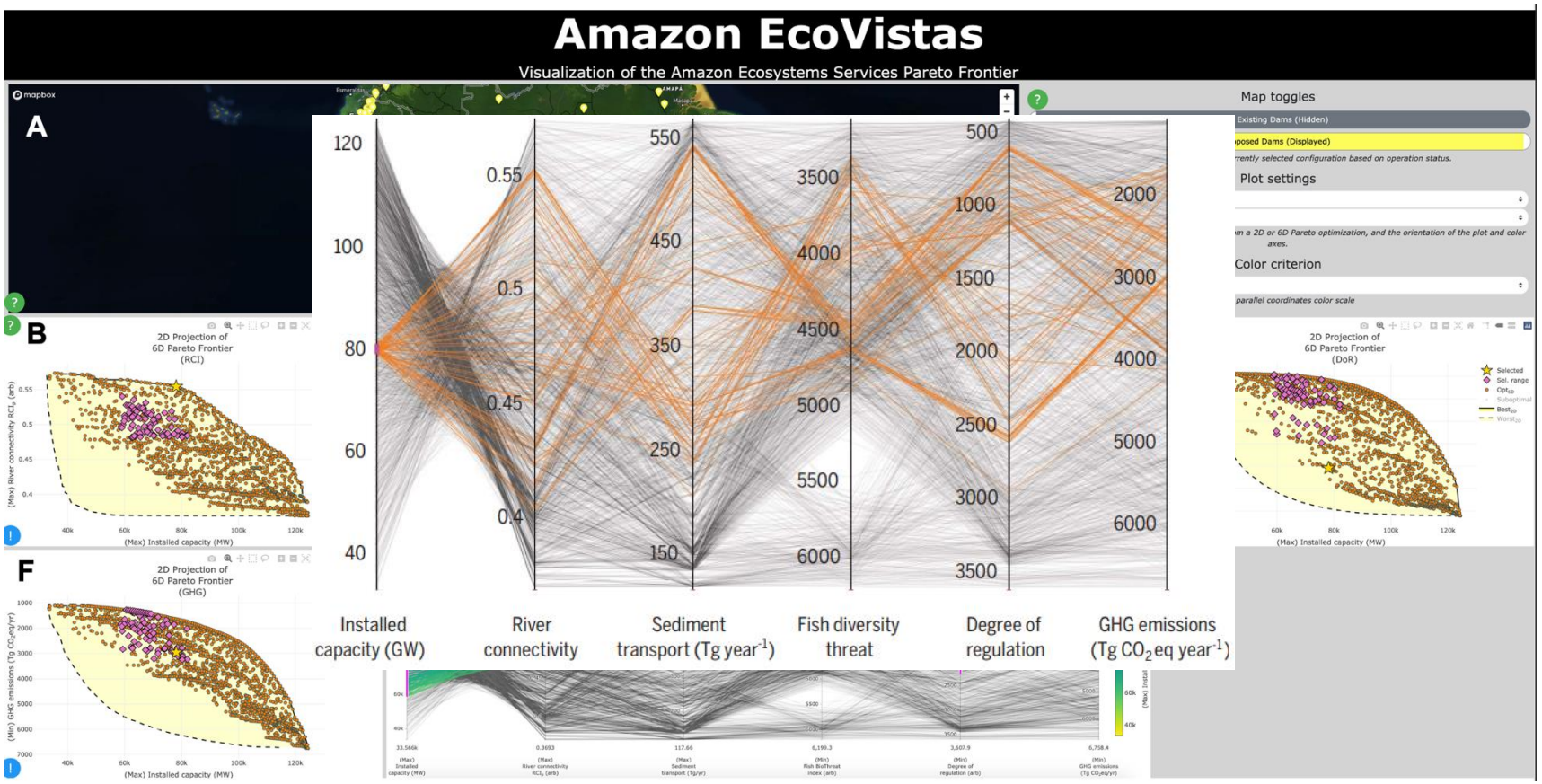
Pareto frontier, optimized with respect to all six criteria projected onto energy and GHG



Marc Grimson, Rafael Almeida, Qinru Shi, Yiwei Bai, Hector Angarita, Felipe Siqueira Pacheco, Rafael Schmitt, Alexander Flecker, Carla P. Gomes. Scaling Up Pareto Optimization for Tree Structures with Affine Transformations: Evaluating Hybrid Floating Solar-Hydropower Systems in the Amazon. AAAI 2024

Visualizing and communicating the Pareto frontier

https://www.cs.cornell.edu/gomes/udiscoverit/amazon-ecovistas

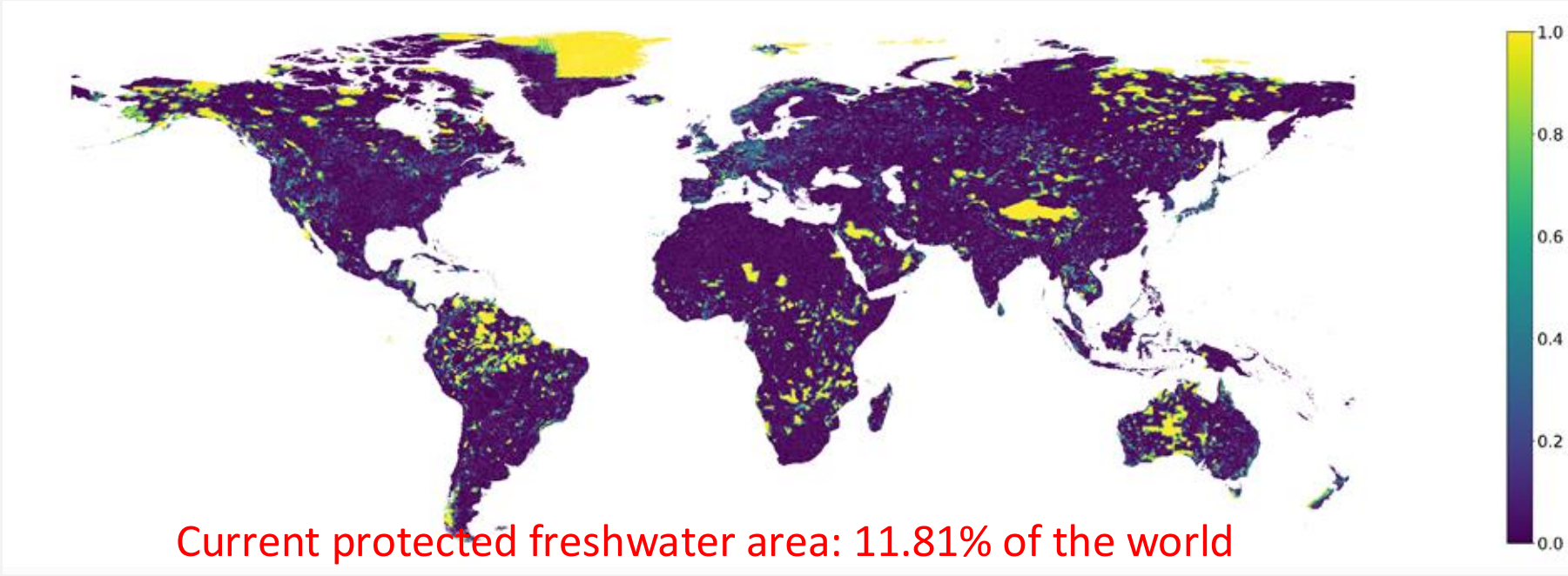


Key challenge!



13,740 freshwater fish species

Global Planning for Freshwater Fishes Under the UN's 30x30 Conservation Goal



Current protected freshwater area: 11.81% of the world

UN's 30x30 Conservation Goal:
Protect and manage 30% of Earth's land, inland waters, and coastal and marine areas by 2030.

Goal: expand current protected areas (PAs) to maximize the protection of freshwater fish species within a 30% area limit.



30% Earth protected by 2030

Inform global priorities for freshwater conservation

WWF TNC IUCN

- Maximize total species protection
- Preserve ecological connectivity
- Enforce 30% area budget

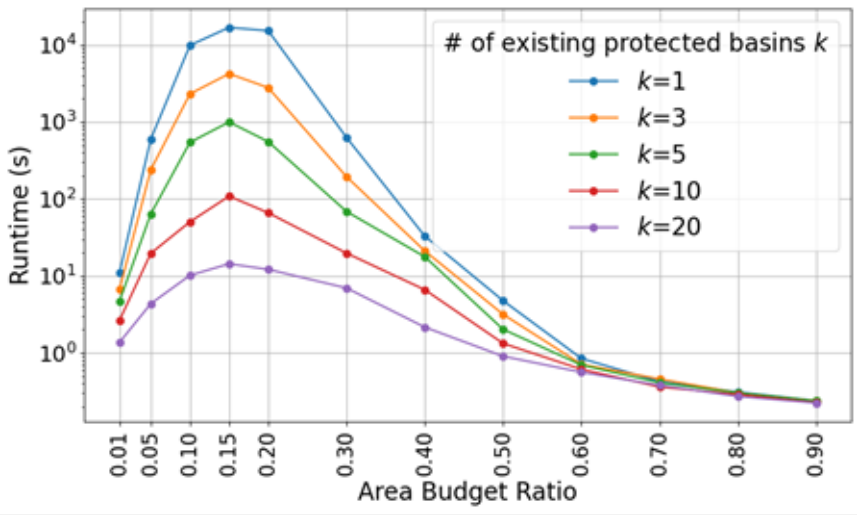
Our contribution:

- Introduced the *Expansion of Connected Components from Alternative Terminals* (ECCAT) Problem (a variant of the Knapsack problem with connectivity constraints) *(IJCAI)*
- Different formulations to solve the ECCAT problem (Mixed Integer Programming)
- Computational and empirical complexity
- A detailed study of the real-world UN 30x30 worldwide freshwater conservation goal

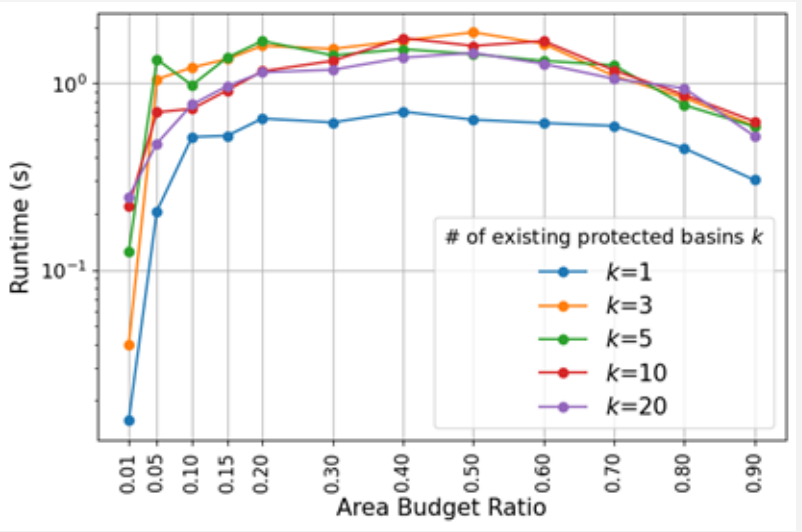
Experimental Results (Synthetic)

Easy-Hard-Easy Complexity Patterns

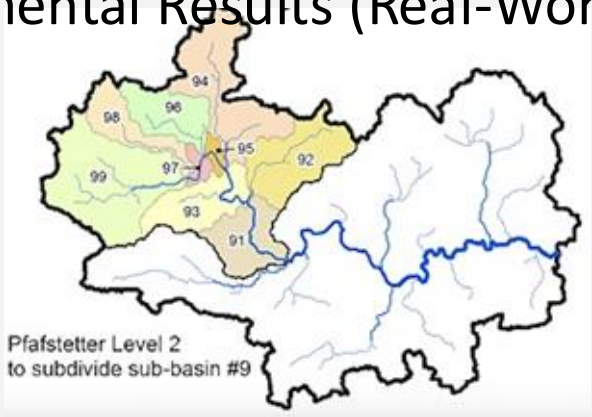
Planar graph (20x20 grid)



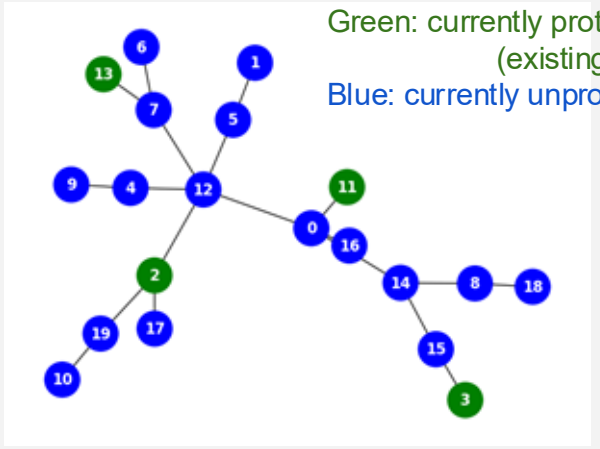
Tree (20x20 star graph)



Experimental Results (Real-World)



River network and sub-basins

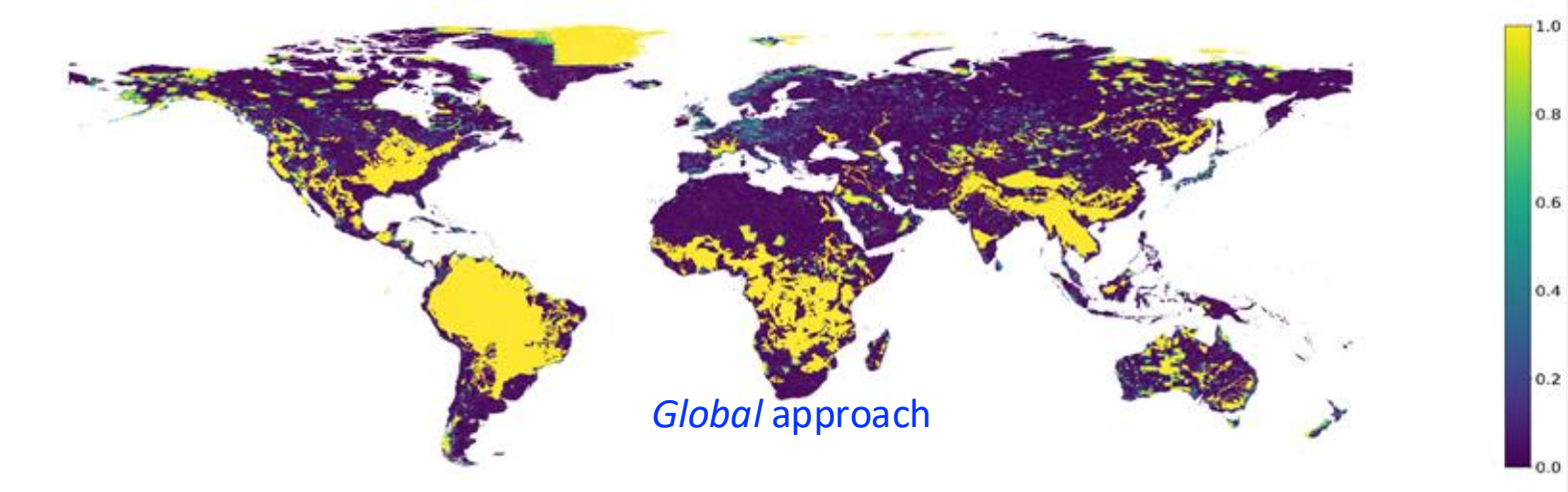


Abstraction as a graph optimization problem

Experimental Results (Real World)

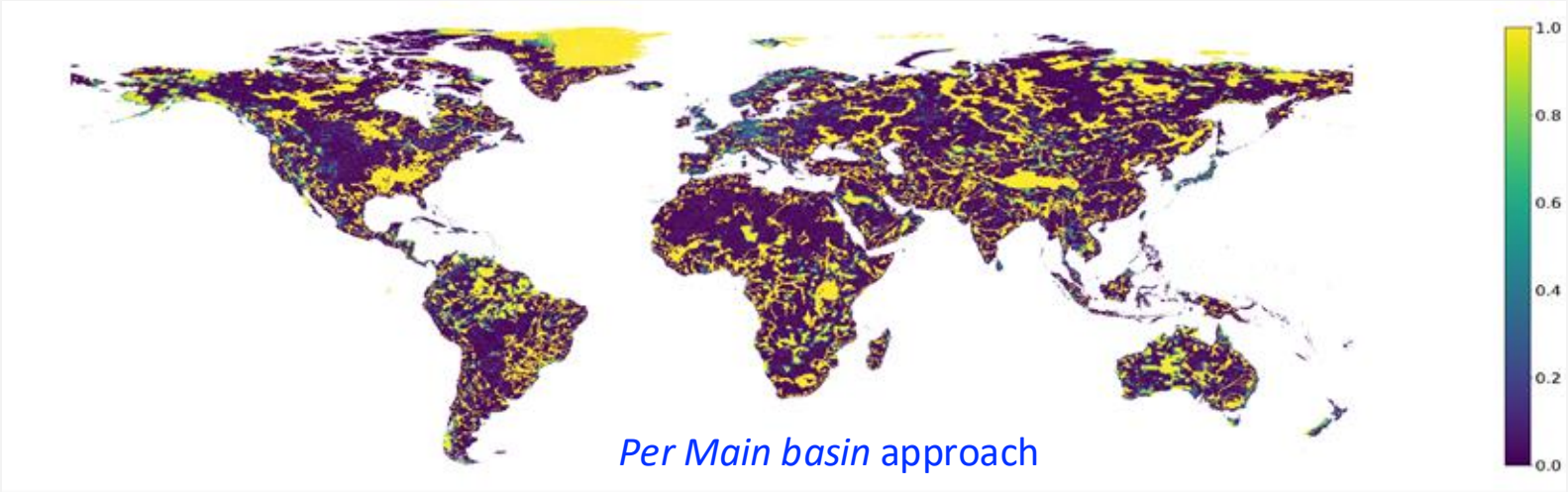
Network-flow-based encoding proved optimality in 1h 20min

(190,675 decision variables)



Global approach

A second approach: optimize per “main basin” (a connected, independent river network, e.g., the Amazon)



Per Main basin approach

Country-level Conservation Commitments

Current



Global approach



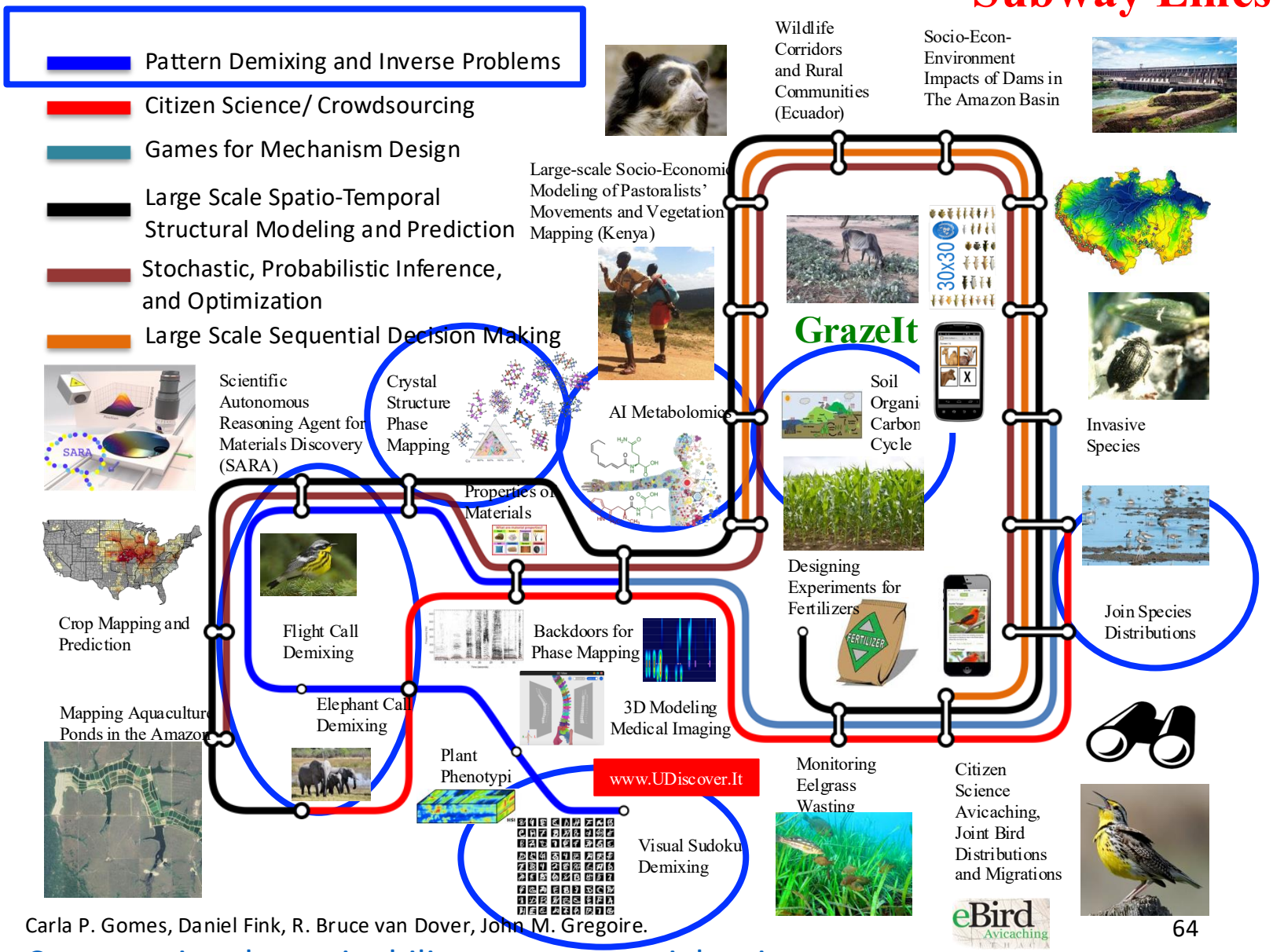
Per Main Basin approach



Question: How to distribute country-level protection?

Transferability of AI Methodologies across domains:
Cross-cutting computational themes

Subway Lines



Carla P. Gomes, Daniel Fink, R. Bruce van Dover, John M. Gregoire.

nature REVIEWS MATERIALS

Computational sustainability meets materials science. *Nature Reviews Materials* (2021).

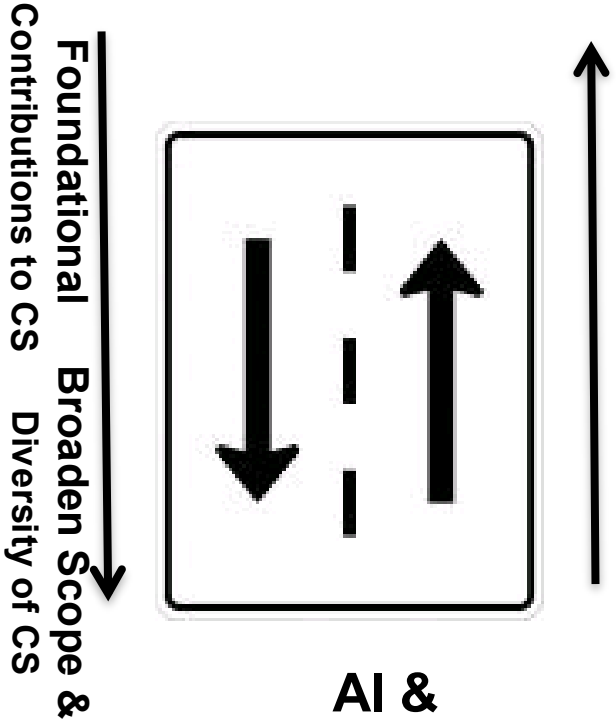
Two Way Street

Scientific Discovery for A Sustainable Future

New challenging problems

New formalisms and concepts

Knowledge-centric AI:
Combining knowledge-centric reasoning,
grounded in first principles, **with data-driven learning**



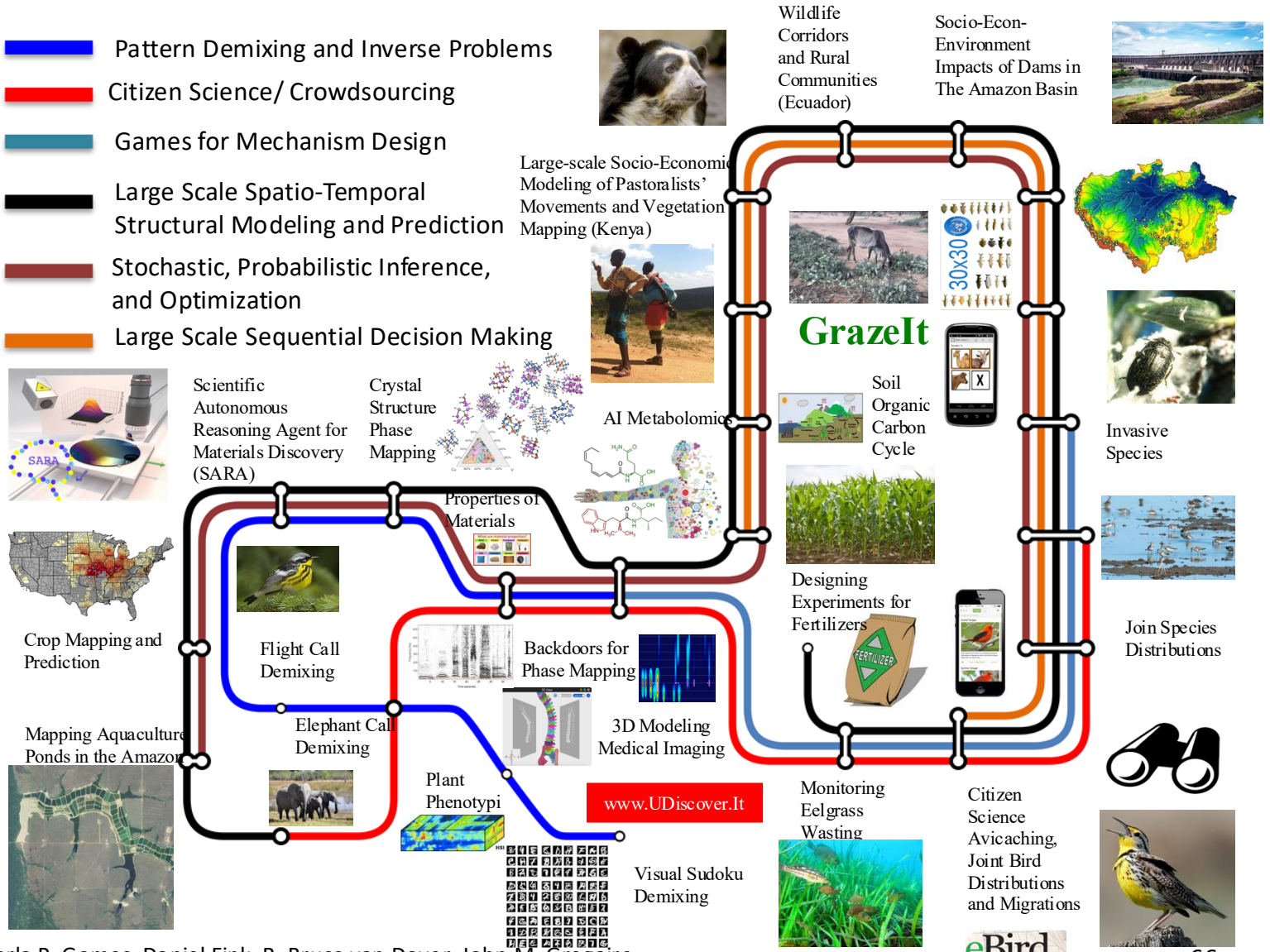
Computational Thinking
providing new
insights,
methodologies,
and solutions

Advancing AI for Humanity

Thank you ☺!

NSF · Air Force Office of Scientific Research · Department of Energy · USDA · NIFA · Army Research Office · Schmidt Sciences

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